第6章　电磁感应

§ 6.1　电磁感应定律

§ 6.2　动生电动势与感生电动势

§ 6.3　互感与自感

§ 6.4　涡电流与趋肤效应

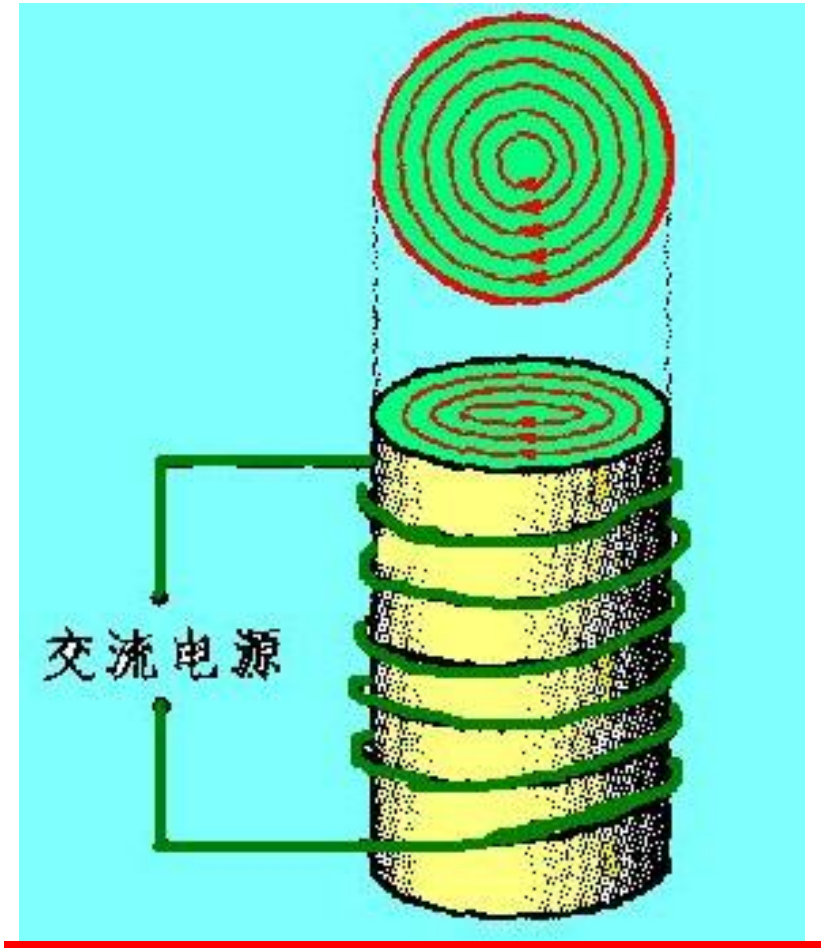
§ 6.5　暂态过程

§ 6.6　磁能

§6.4 涡电流与趋肤效应

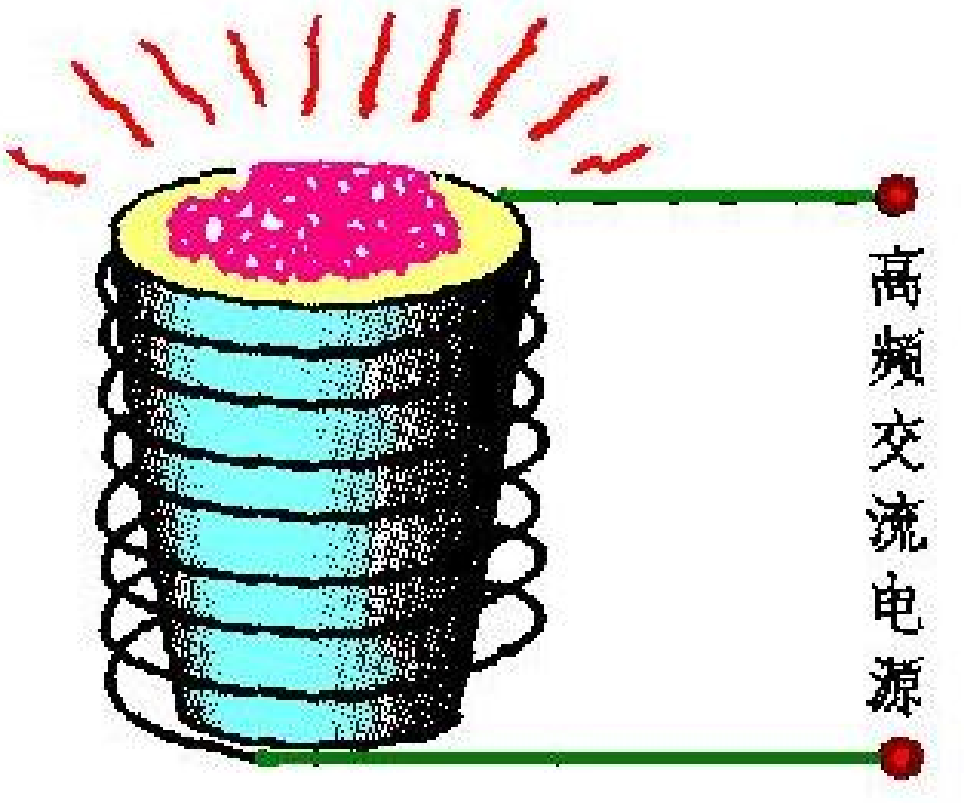
1. 涡电流

- 1855年傅科发现金属块处在变化的磁场中或相对于磁场运动时，其内部会产生感应电流。
- 铁芯可看作是由一系列半径逐渐变化的圆柱状薄壳组成，每层薄壳自成一个闭合回路，在交变磁场中，通过这些薄壳的磁通量都在不断地变化，所以一层层地产生感应电流。

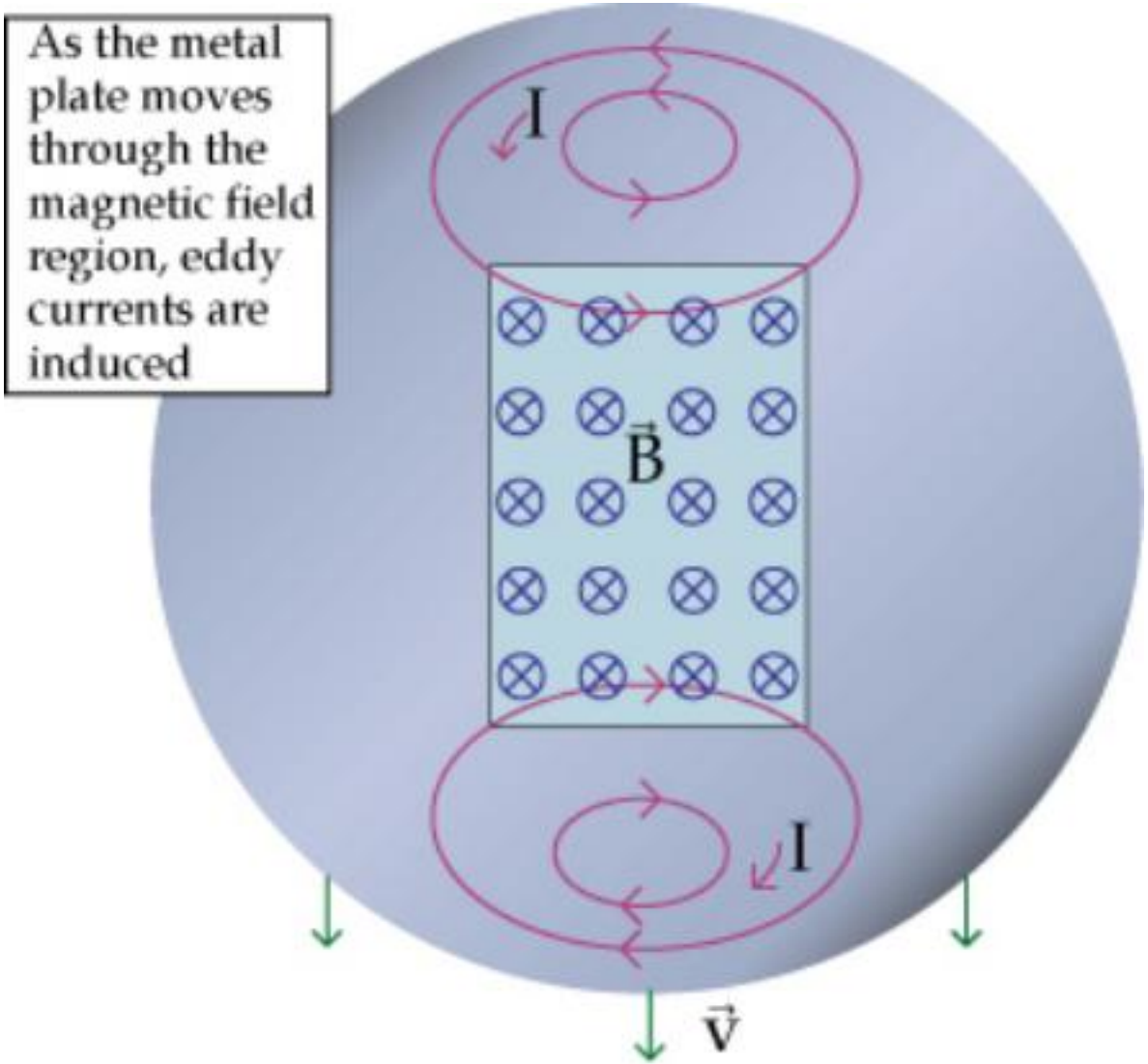


傅科电流(Eddy current)
1855年获科普利奖

- 从上端俯视，电流流线呈闭合的涡旋状，因而这种感应电流叫做涡电流．简称为涡流。由于大块金属的电阻很小，因此涡流可达非常大的强度。



高频电磁感应炉

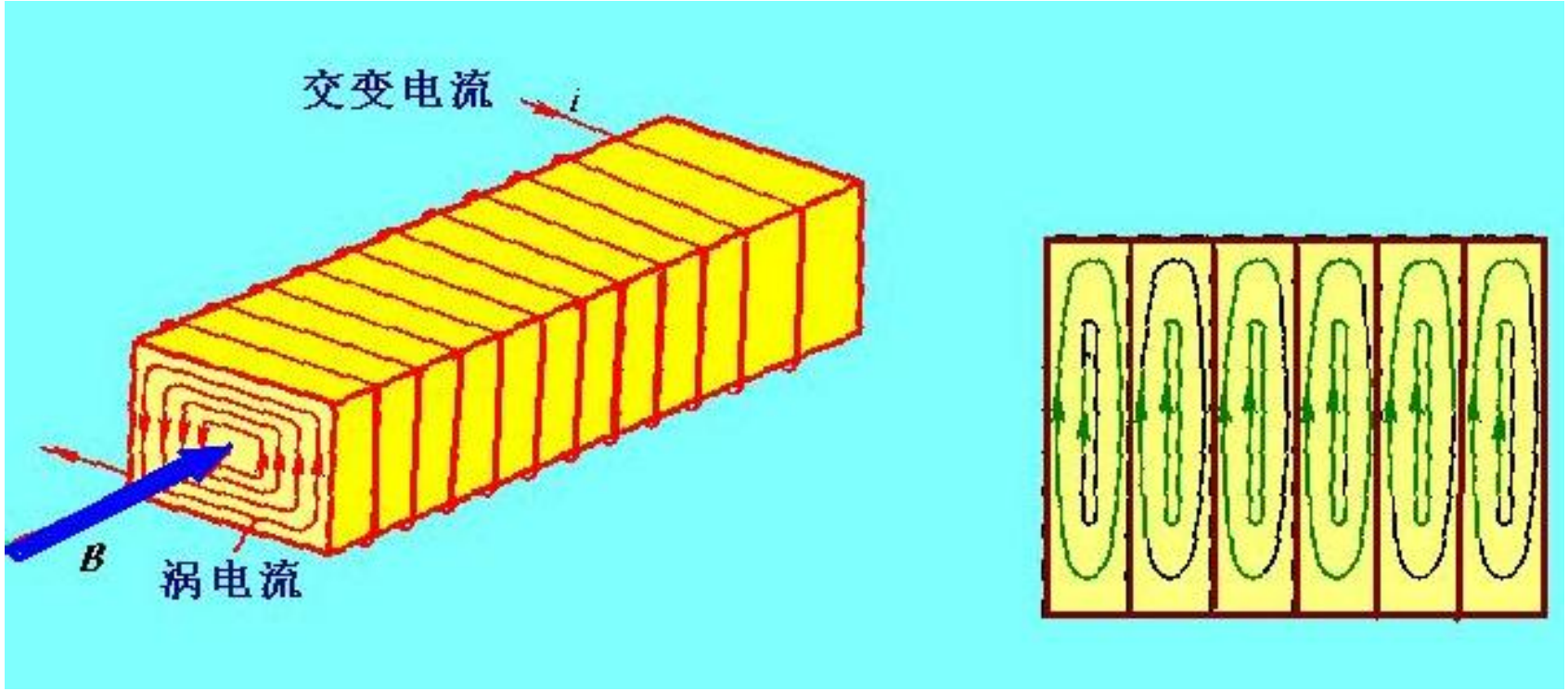


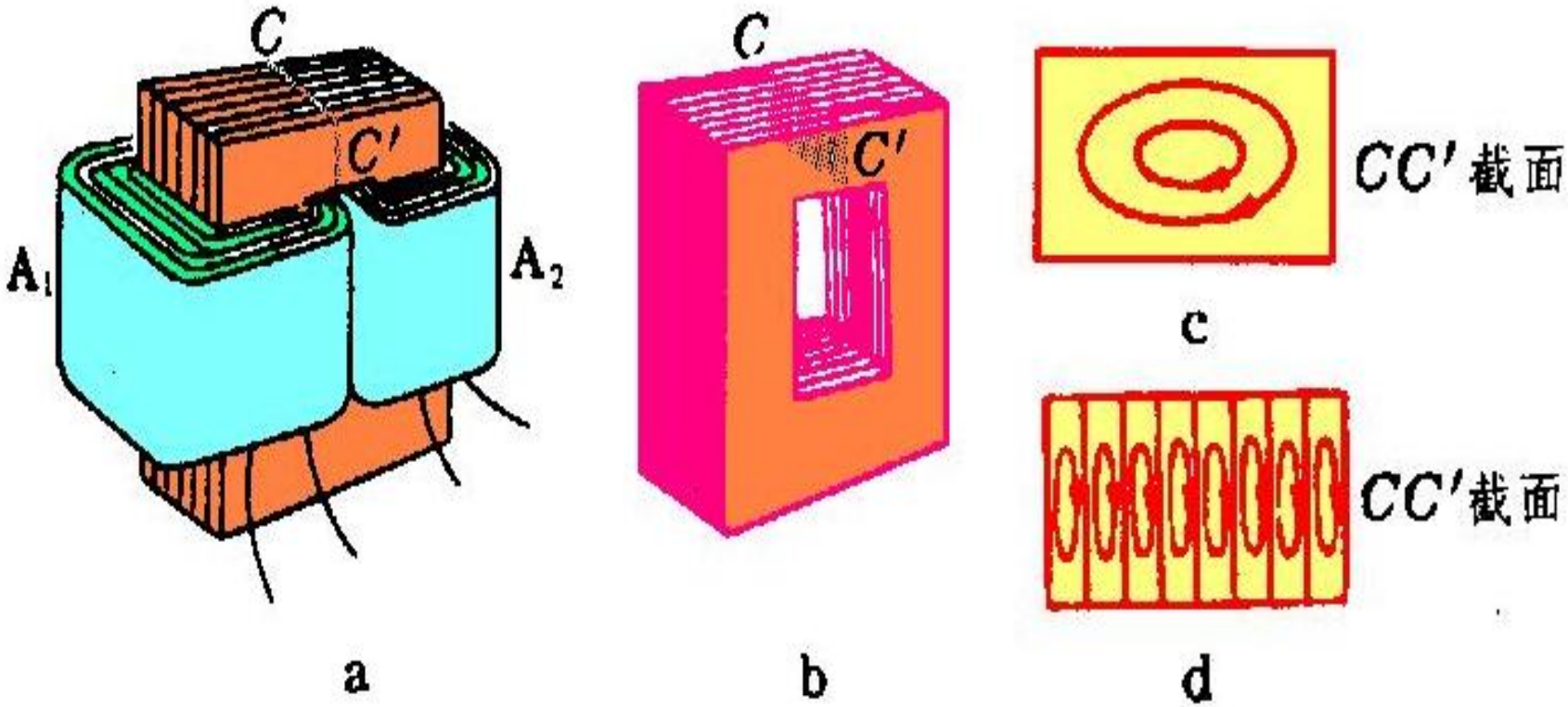
涡电流（**Eddy Current**）的方向可以通过楞次定律来判定。

✹减少涡电流的方法

（1）高电阻材料，硅钢，在钢中增加硅，而磁导率与铁差不多。

（2）多层绝缘片叠加而成，减少涡电流的导体截面积。

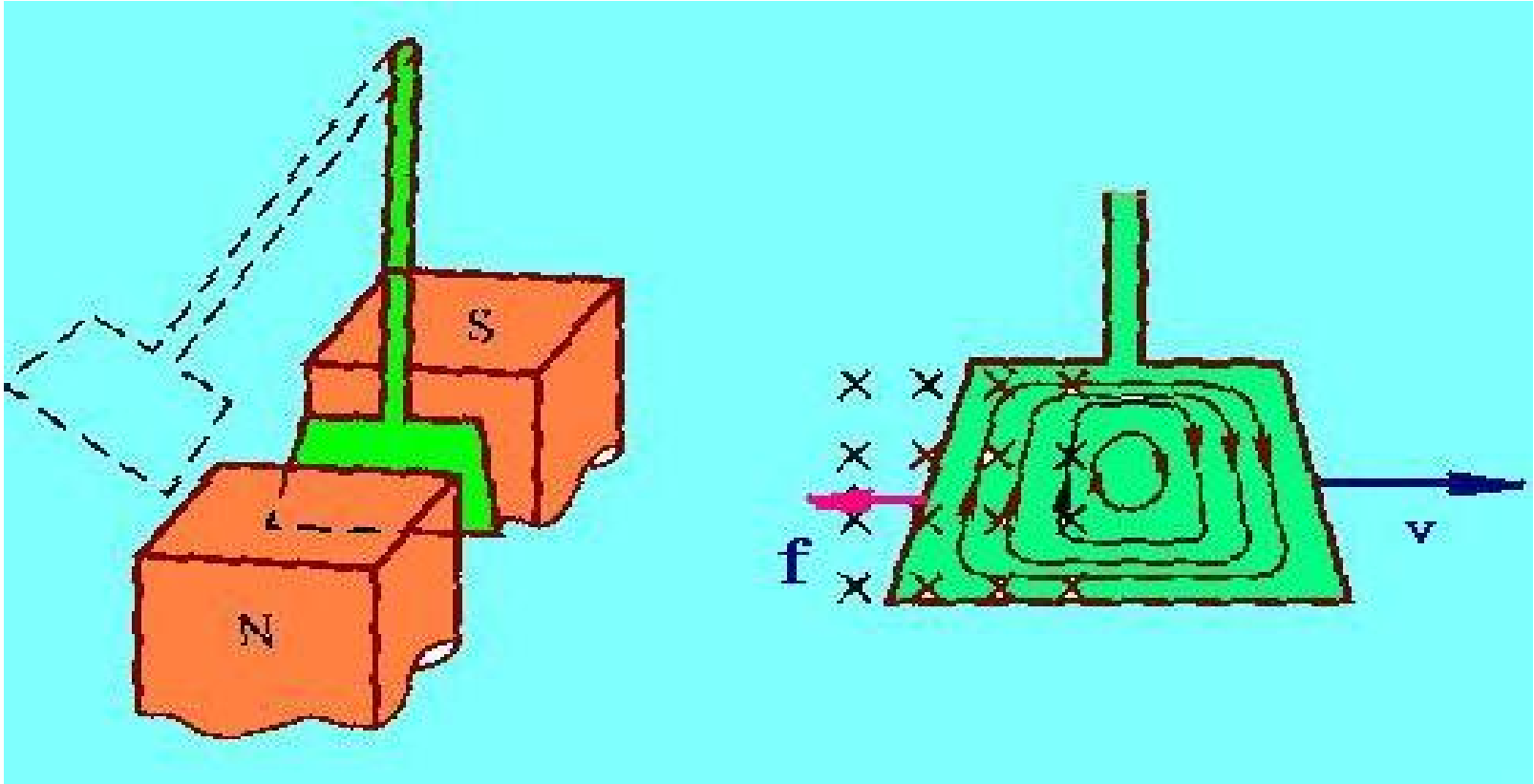




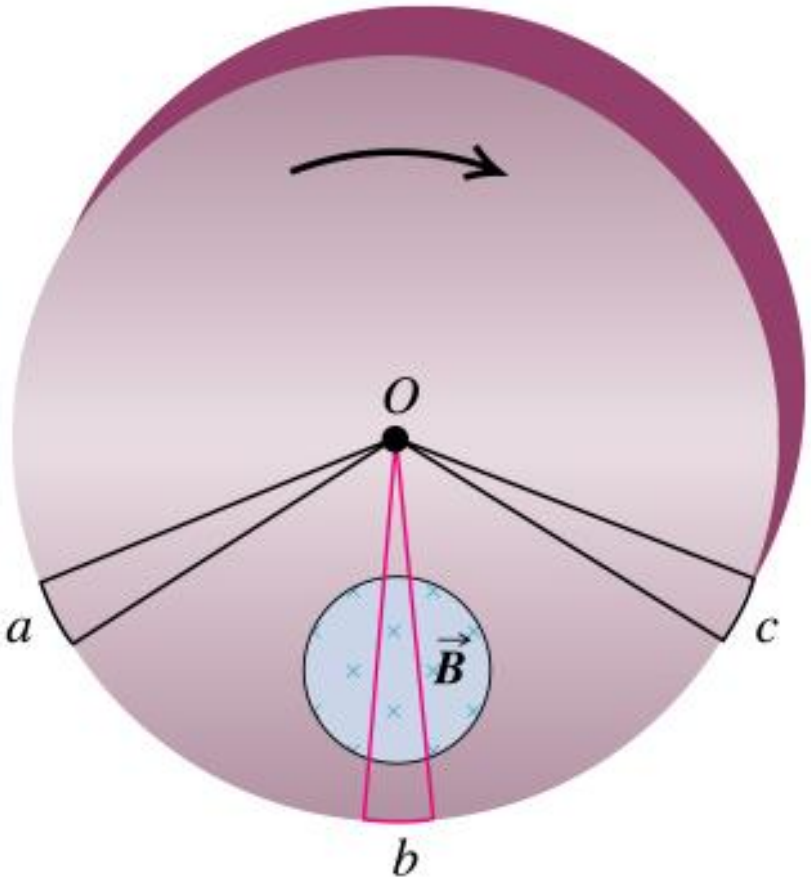
变压器的铁芯

2. 电磁阻尼

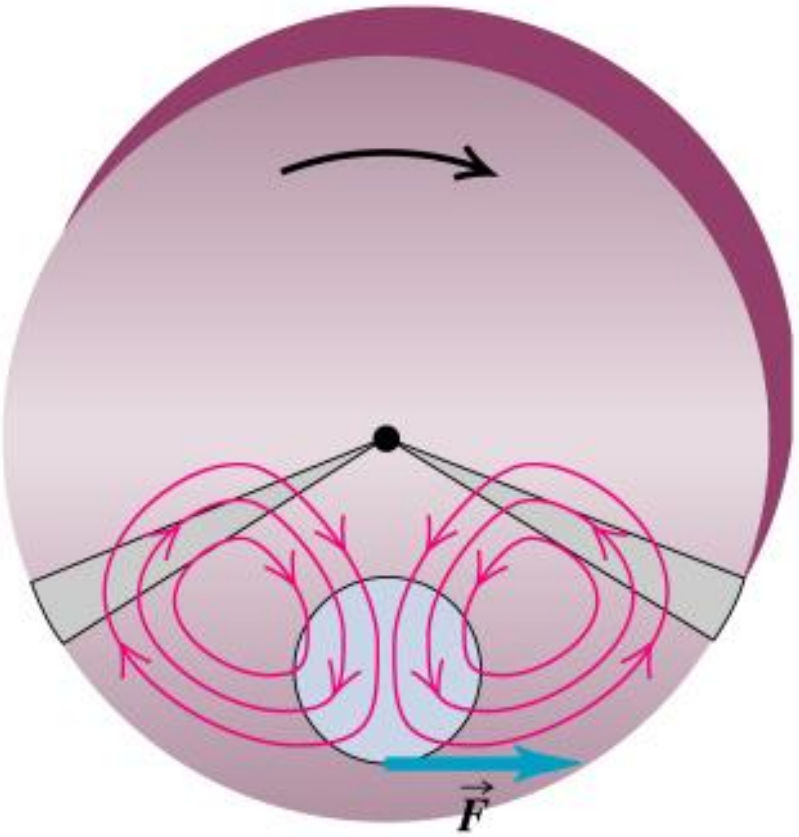
涡电流除了热效应外，还有机械效应，大块金属在磁场中运动会受到很大的阻力。



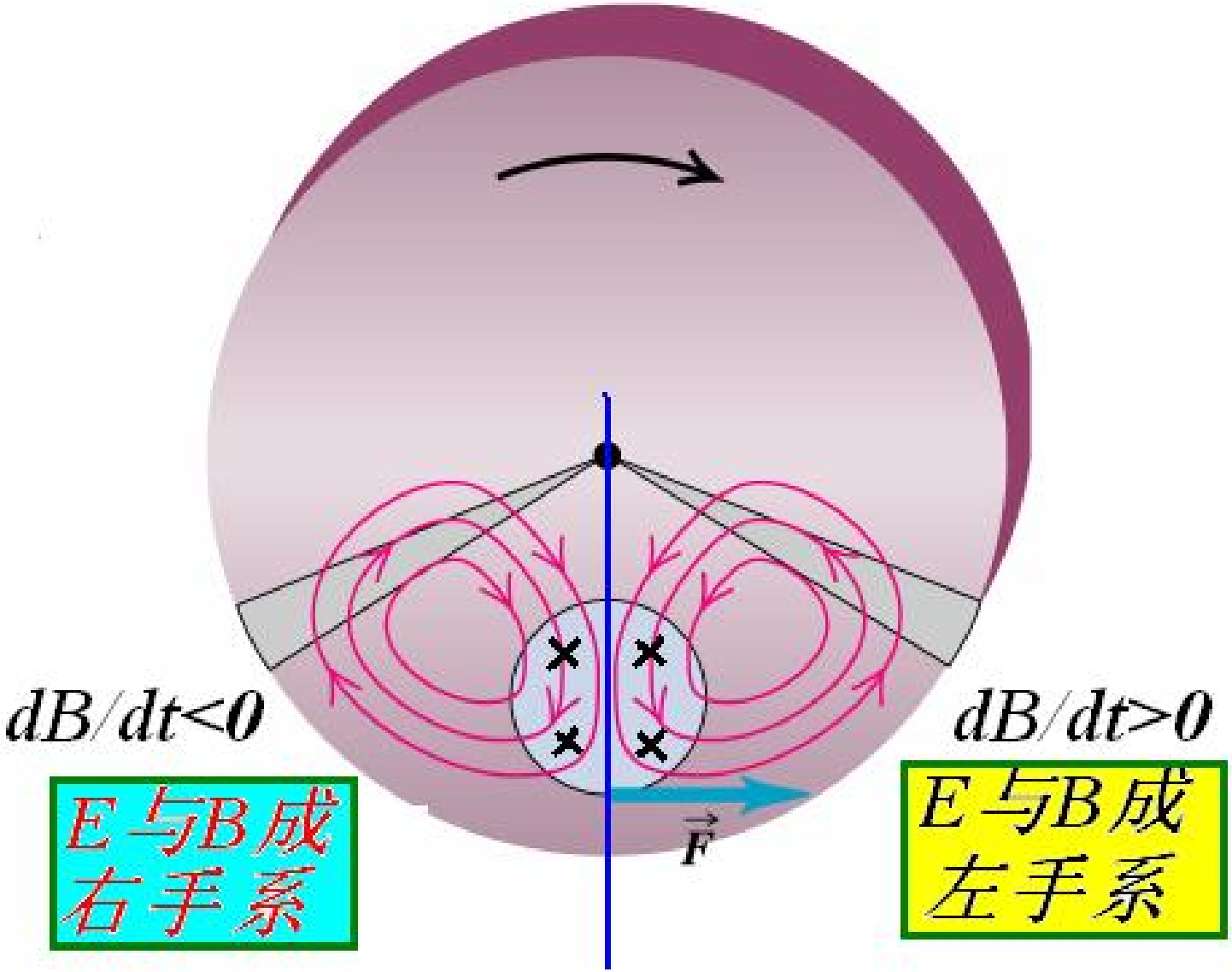
电磁制动

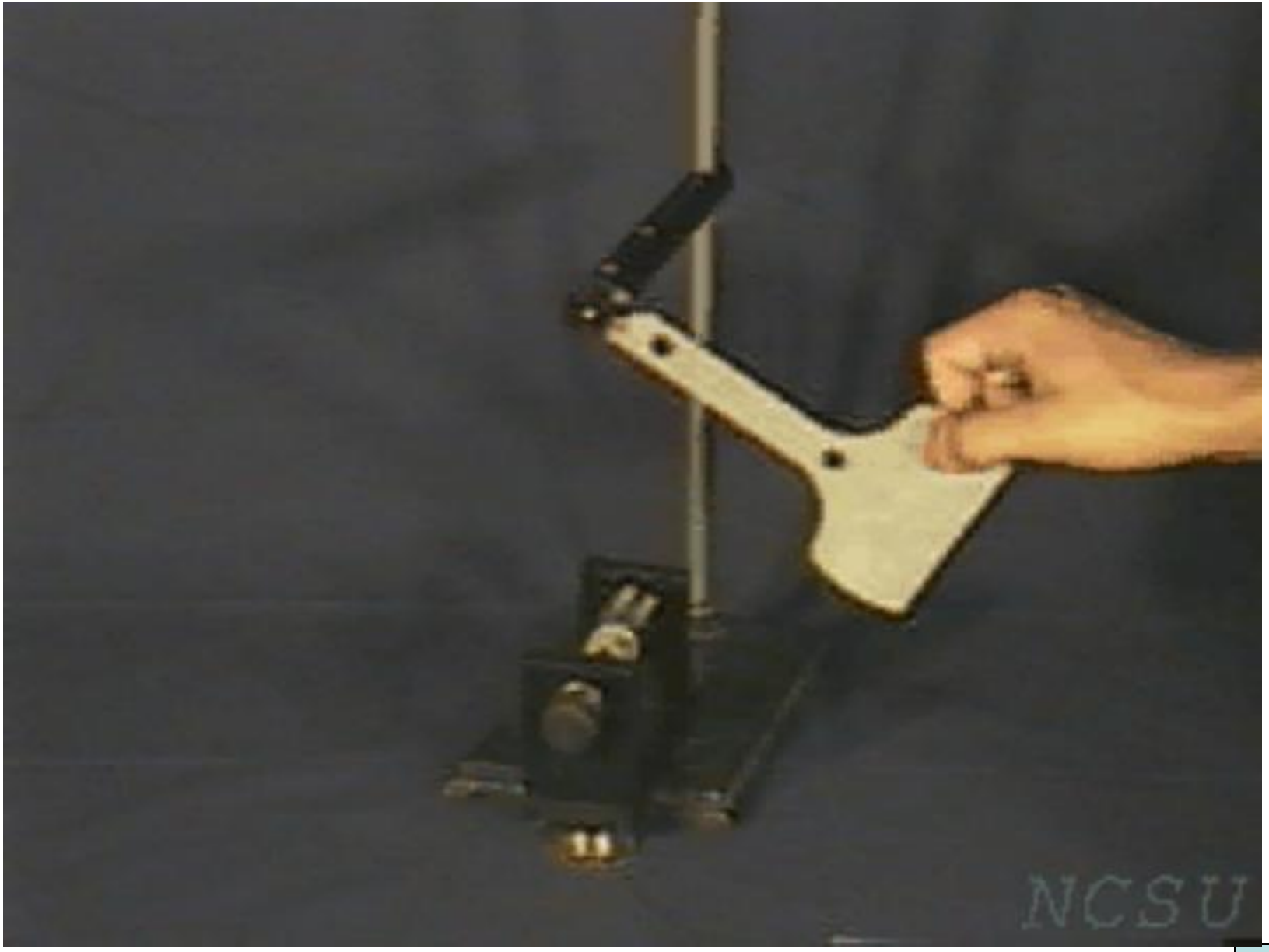


(a)



(b)

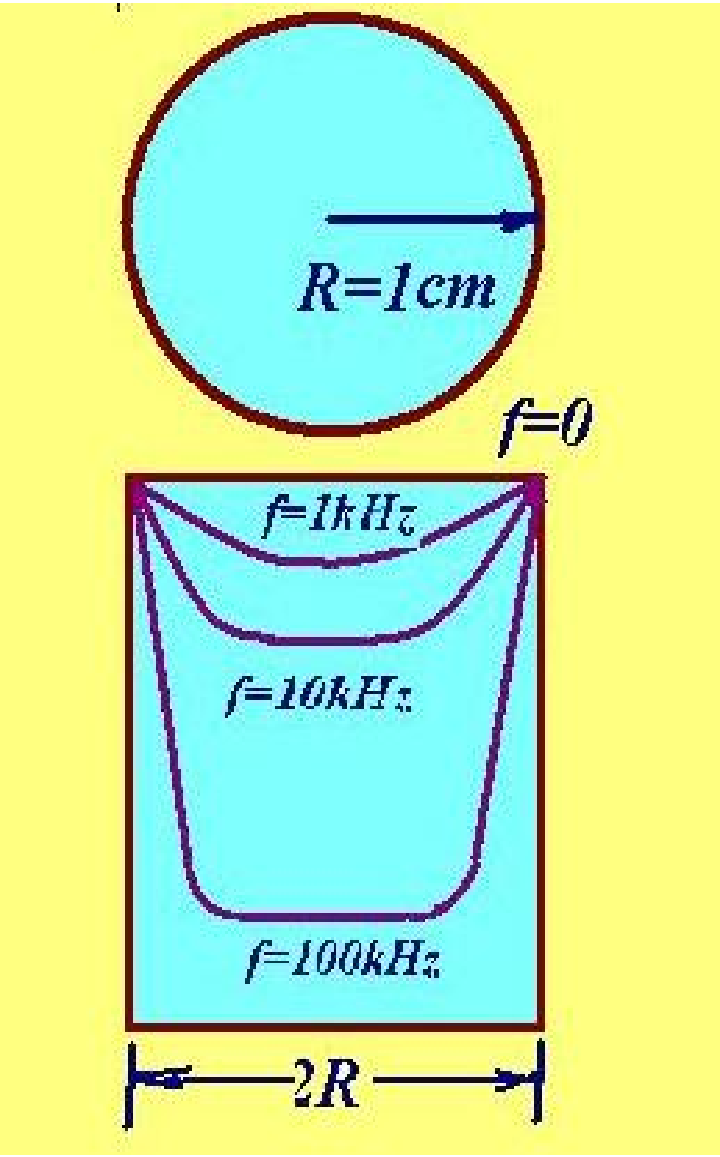

dB/dt<0
E与B成
右手系
dB/dt>0
E与B成
左手系
$\vec{F}$


NCSU

趋肤效应

1. 趋肤效应

- 当交变电流通过柱体时，电流密度分布不再均匀，越靠近导体表面处，电流密度越大，称趋肤效应。
- 其结果使有效面积减少，电阻增加。
- 通常采用细导线编织成束代替实心导线。
- 应用：表面淬火，表面变硬。



*2. 趋肤效应的机制

导体内：

$$\varepsilon \nabla \cdot \vec{E} = \rho$$

$$\vec{j} = \sigma \vec{E}$$

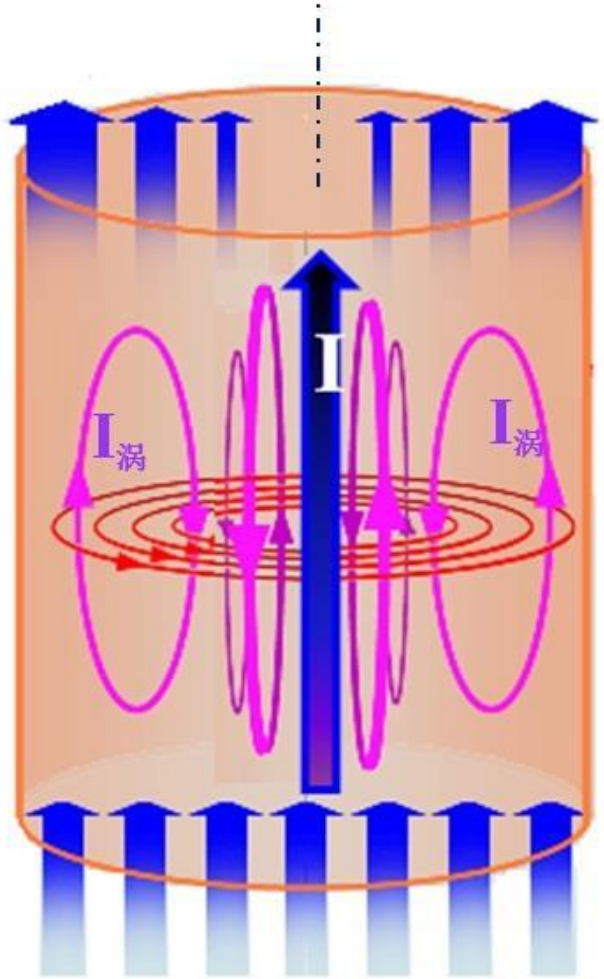
$$\nabla \cdot \vec{j} = \frac{\sigma}{\varepsilon} \rho$$

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot \vec{j} = -\frac{\sigma}{\varepsilon} \rho$$

解之：

$$\rho(t) = \rho_0 e^{-\frac{\sigma}{\varepsilon} t} = \rho_0 e^{-\frac{t}{\tau}}$$

其中 $\tau = \frac{\varepsilon}{\sigma}$



当$\omega << \frac{1}{\tau} = \frac{\sigma}{\varepsilon}$

或$\frac{\sigma}{\varepsilon\omega} >> 1$

$$\rho(t) = 0$$

良导体内无自由电荷分布，电荷只能分布在导体表面。

$$\vec{E} = E_0 e^{-\alpha z} e^{i(\beta z - \omega t)}$$

$$j = j_0 e^{-d/d_s}$$

d_s为趋肤深度。

$$d_s = \sqrt{\frac{2}{\omega\mu_0\mu\sigma}}$$

铜：$\sigma = 5.9\times10^7(\Omega m)^{-1}, \mu = 1, f = 1kHz$

$$d_s = 0.21cm$$

若$f = 100kHz,\ d_s = 0.021cm$

对铁，由于μ很大，趋肤效应更明显。

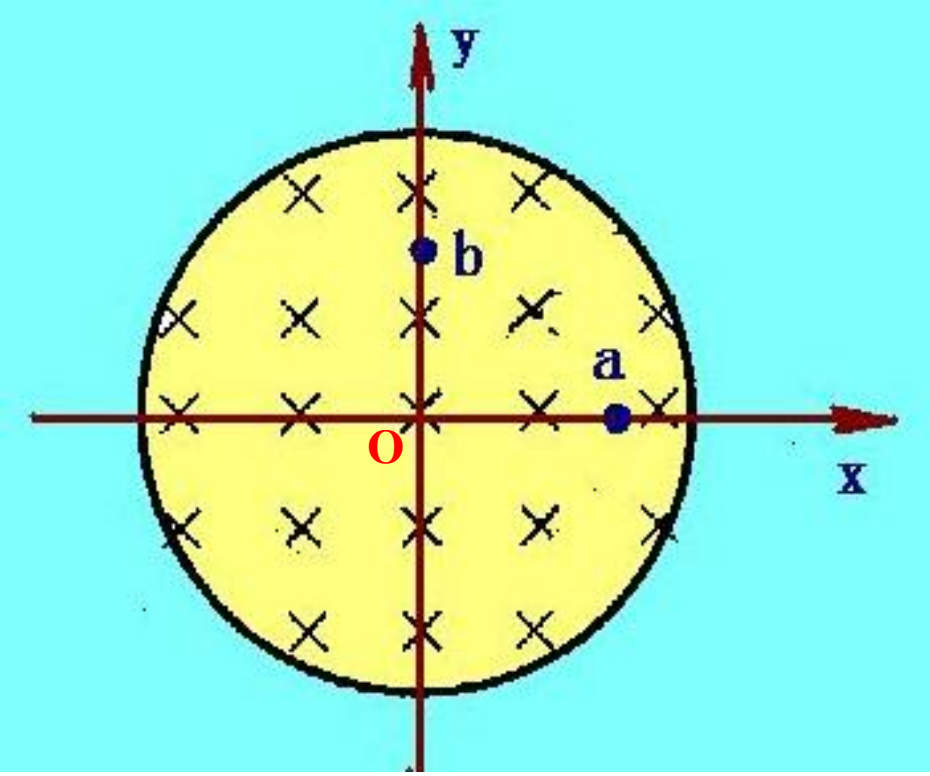
例 题

[例]如图，磁感应强度B在圆内均匀分布，且$B = B_0 \sin\omega t$，求a，b两点间的电势差。（ob = oa = r）

（1）ab之间用跨过第2，3，4象限的圆弧导线连接；

（2） ab之间用跨过第1象限的圆弧导线连接；

（3） ab之间没有导线连接；



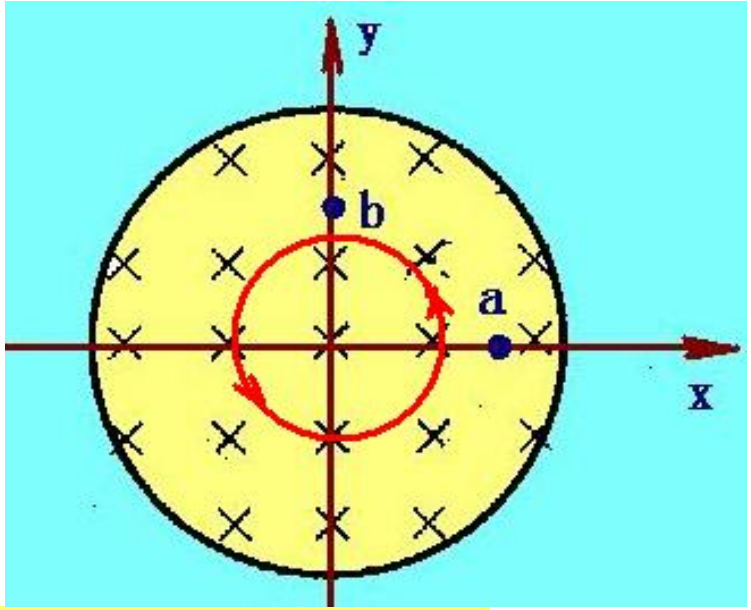
[解]

$$\oint_L \vec{E}_{旋} \cdot d\vec{l} = -\iint \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$$

对半径为r的顺时针圆形回路

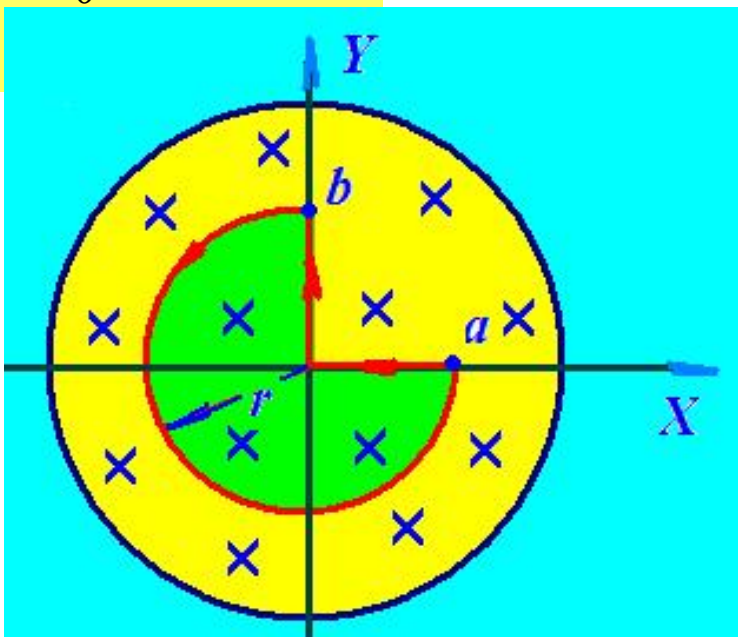
$$\vec{E}_{旋} 2\pi r = -(+\frac{dB}{dt}) \quad \pi r^2 = -B_0 \omega \cos \omega t \cdot \pi r^2$$

$$\therefore E_{旋} = -\frac{1}{2} B_0 \omega r \cos \omega t$$

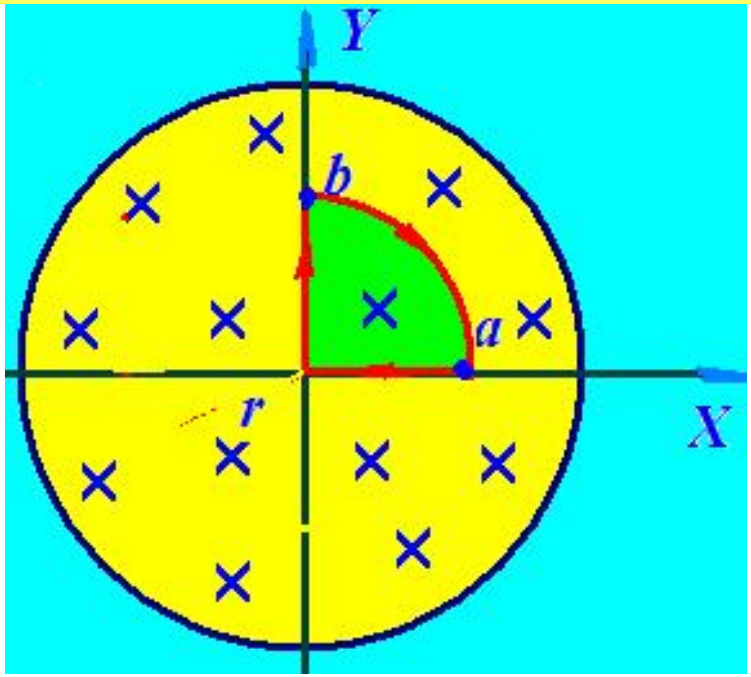


负号表示E旋转方向为逆时针方向

$$(1)\ V_{ab} = \varepsilon = \int_{b(2\to3\to4)}^{a} \vec{E}_{旋} \cdot d\vec{l} = \frac{3}{4}\pi r^2 B_0 \omega \cos \omega t$$



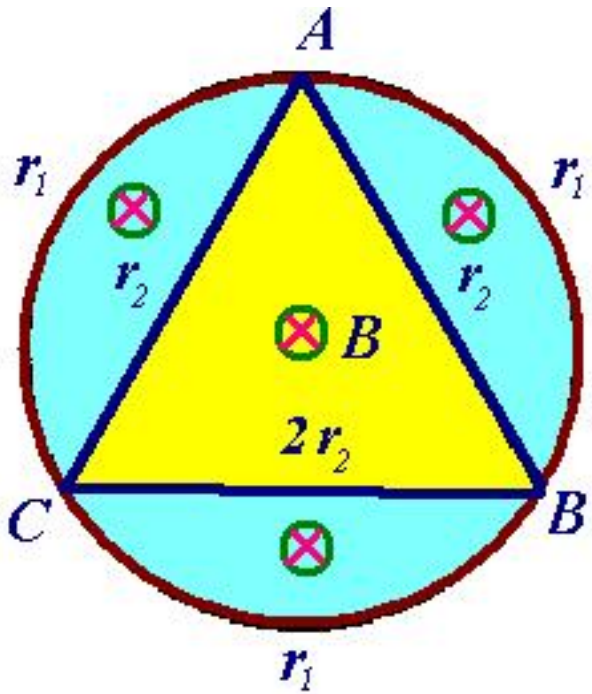
$$(2)\ V_{ab}=\varepsilon=\int_{b(1)}^{a}\vec{E}_{旋}\cdot d\vec{l}=-\frac{1}{4}\pi r^2 B_0\omega\cos\omega t$$

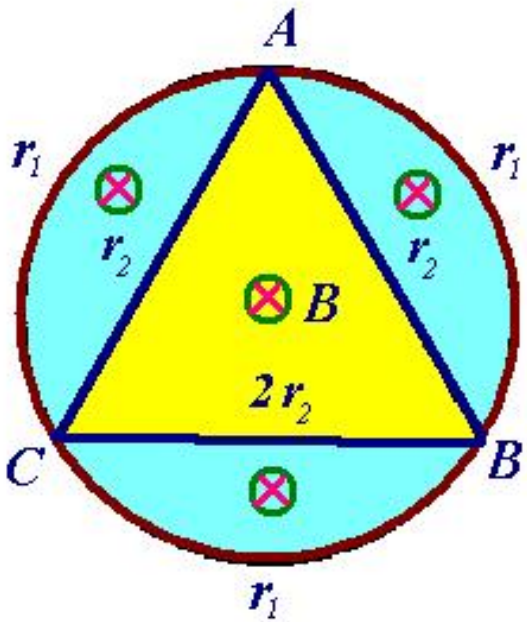


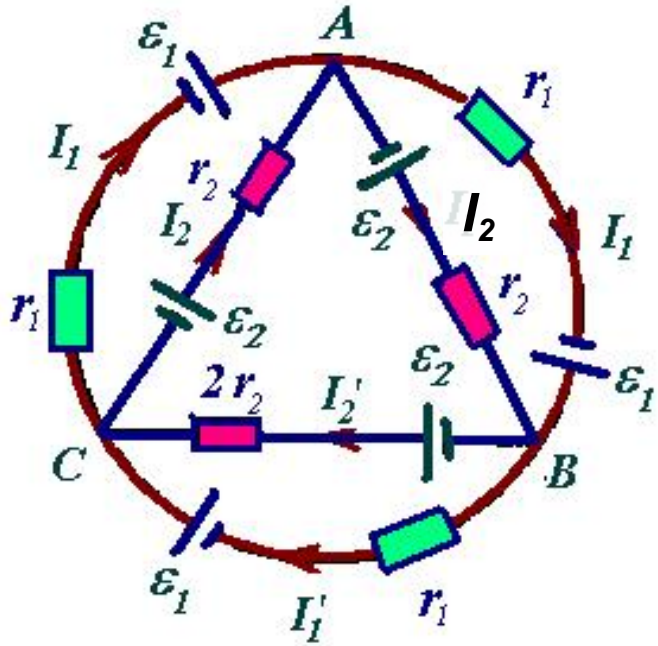
$$(3)\ V_{ab}=0(\text{没有空间电荷分布})$$

即电动势由非静电力产生，它与积分路径有关，且对闭合回路可以不等于零。谈两点间的电动势无意义，只能说***ab路径***的感应电动势，它与ab路径的几何形状有关。

[例题]如图所示，由均质细导线弯成的半径为a的圆线圈和一内接等边三角形的电阻丝构成的电路，电路中各段电阻值已在图中标明．在圆线圈所在平面内有垂直纸面向里的匀强磁场B，B随时间t均匀地减小，其变化率的大小为常量k，设$2r_1=3r_2$． 试求图中A、B两点之间的电势差U。







[分析]磁场随时间变化，导致磁通量变化，使得圆线圈和三角形电阻丝内分别有感应电动势ε_c和ε_t，由于对称性，三条圆弧各有电动势$\varepsilon_1=(1/3)\varepsilon_c$，三角形的三边各有电动势$\varepsilon_2=(1/3)\varepsilon_t$．于是可将图中的电路改画为右图，再根据对称性标明各支路的电流大小和方向．利用直流电路的基尔霍夫定律即可求解。

根据B的方向及均匀减小，可知圆周上感应电动势的方向为顺时针，大小为：

$$\varepsilon_C = -\frac{\Delta\Phi_C}{\Delta t} = -\frac{\Delta}{\Delta t}(B\pi a^2) = k\pi a^2$$

三角形周边上的感应电动势的方向也是顺时针，大小为：

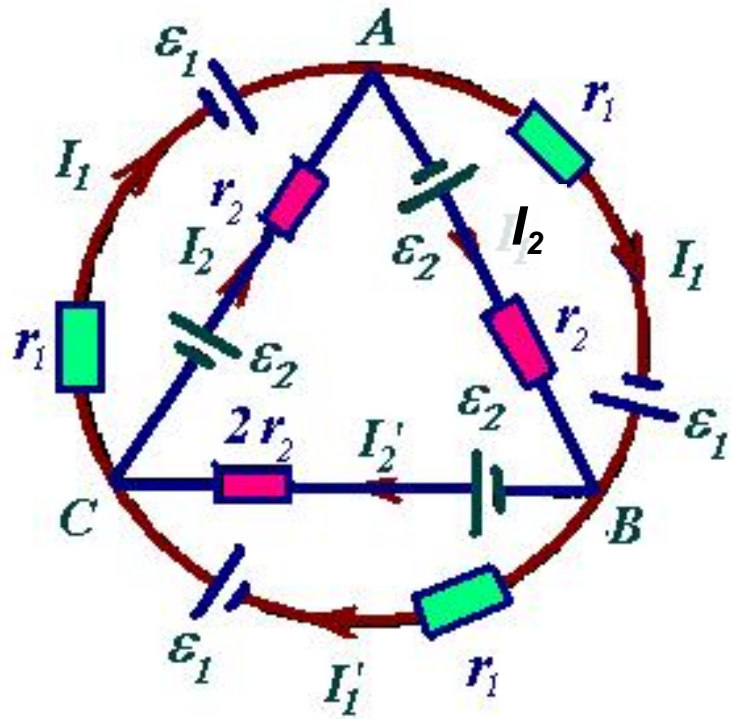
$$\varepsilon_t = -\frac{\Delta\Phi_t}{\Delta t} = -\frac{\Delta}{\Delta t}(B\cdot\frac{1}{2}\cdot\frac{3}{2}a\cdot\sqrt{3}a) = -\frac{3}{4}\sqrt{3}a^2\frac{\Delta B}{\Delta t} = \frac{3}{4}\sqrt{3}ka^2$$

因对称性，ε_c可等分为三段圆弧上的三个ε_1，大小为：

$$\varepsilon_1 = \frac{1}{3}\varepsilon_c = \frac{1}{3}k\pi a^2$$

同样ε_t可等分为三角形三边上的三个ε_2，大小为：

$$\varepsilon_2 = \frac{1}{3}\varepsilon_t = \frac{\sqrt{3}}{4}ka^2$$



电路中电源、电阻和电流的分布如图所示，由基尔霍夫定律，有：

$$\begin{cases} I_1 + I_2 = I_1' + I_2' \\ 2I_1r_1 + I_1'r_1 - 3\varepsilon_1 = 0 \\ 2I_2r_2 + 2I_2'r_2 - 3\varepsilon_2 = 0 \\ I_1r_1 - I_2r_2 - \varepsilon_1 + \varepsilon_2 = 0 \end{cases}$$

已知：

$$r_2 = \frac{2}{3} r_1$$

把各电动势值代入方程，有：

$$\begin{cases} I_1' + I_2' = I_1 + I_2 \\ 2I_1 + I_1' = \dfrac{k\pi a^2}{r_1} \\ I_2 + I_2' = \dfrac{9\sqrt{3}ka^2}{16r_1} \\ 3I_1 - 2I_2 = (\pi - \dfrac{3\sqrt{3}}{4})\dfrac{ka^2}{r_1} \end{cases}$$

消去I_1'和I_2'，得：

$$3I_1 + 2I_2 = (\pi + \frac{9}{16}\sqrt{3})\frac{ka^2}{r_1}$$

解之，得：

$$I_1 = \left(\frac{\pi}{3} - \frac{\sqrt{3}}{32}\right)\frac{ka^2}{r_1}$$

故A, B两点的电势差为：

$$U = I_1 r_1 - \varepsilon_1 = -\frac{\sqrt{3}}{32}ka^2$$

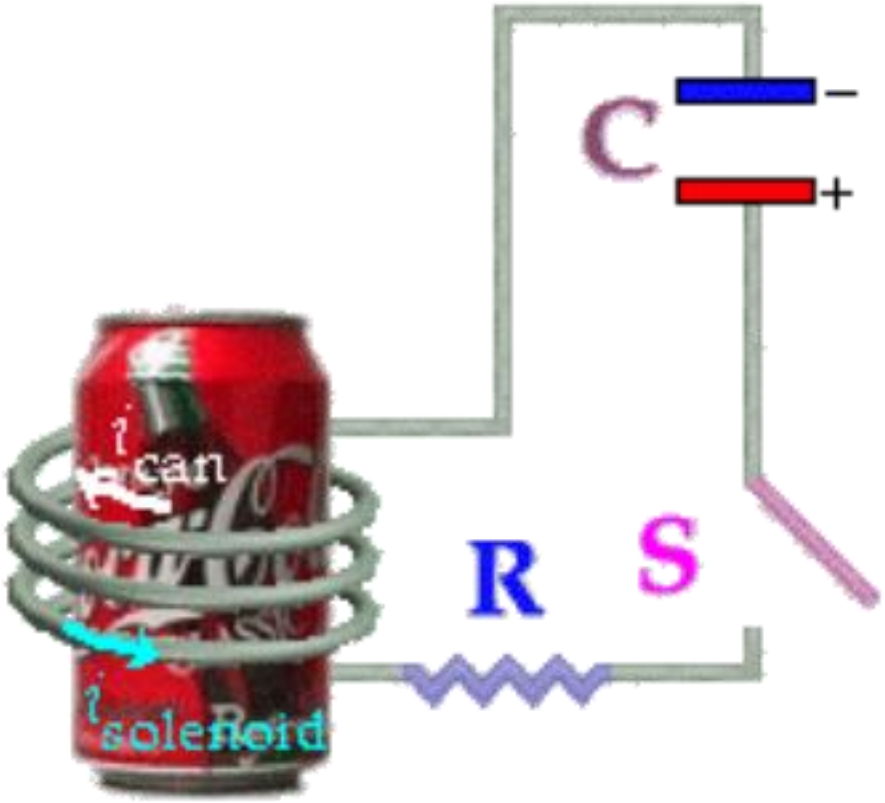
解题注意

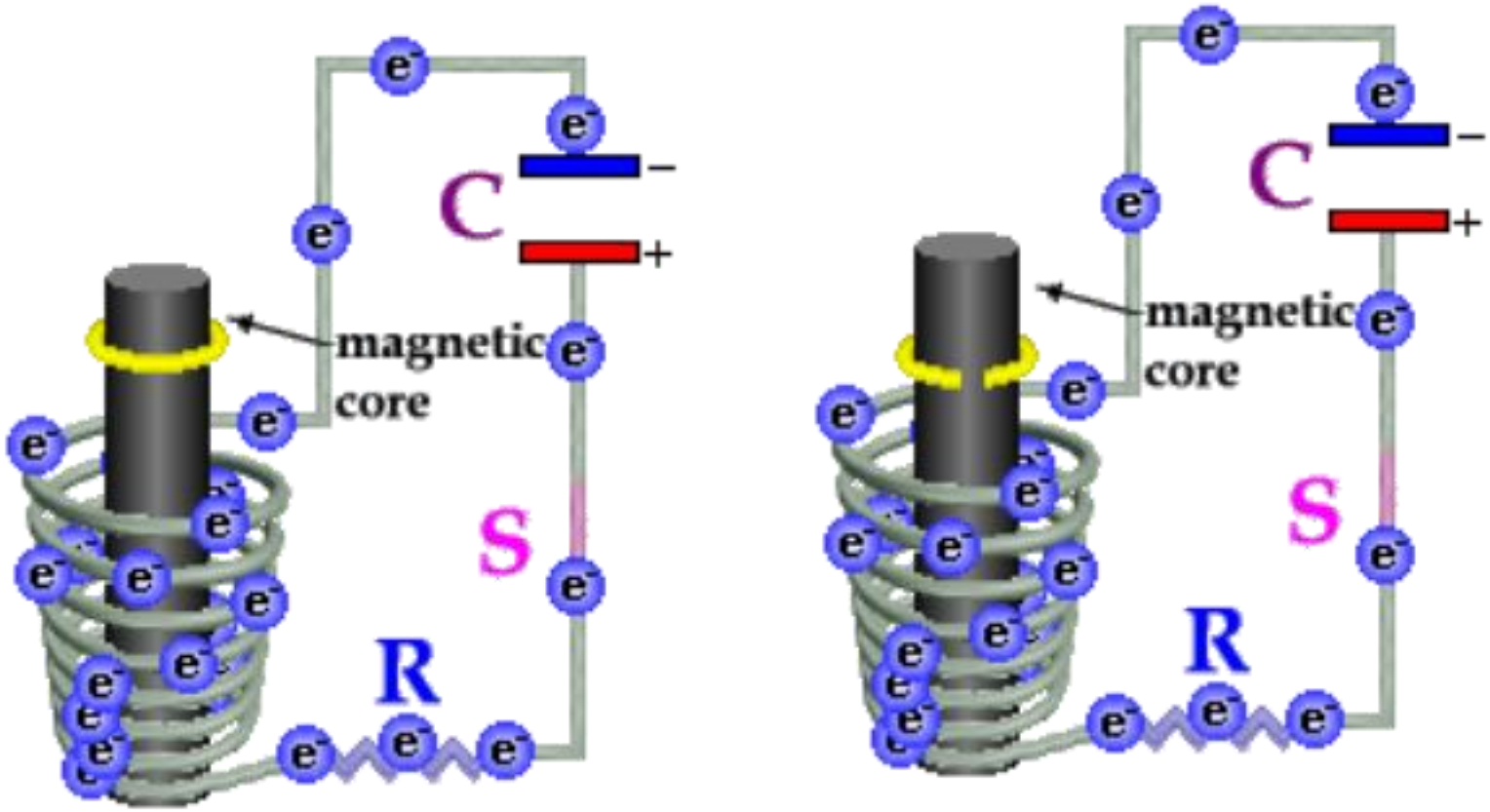
(1) 电动势的方向

(2) 电动势的大小

(3) 结合基尔霍夫定律

电磁感应的应用

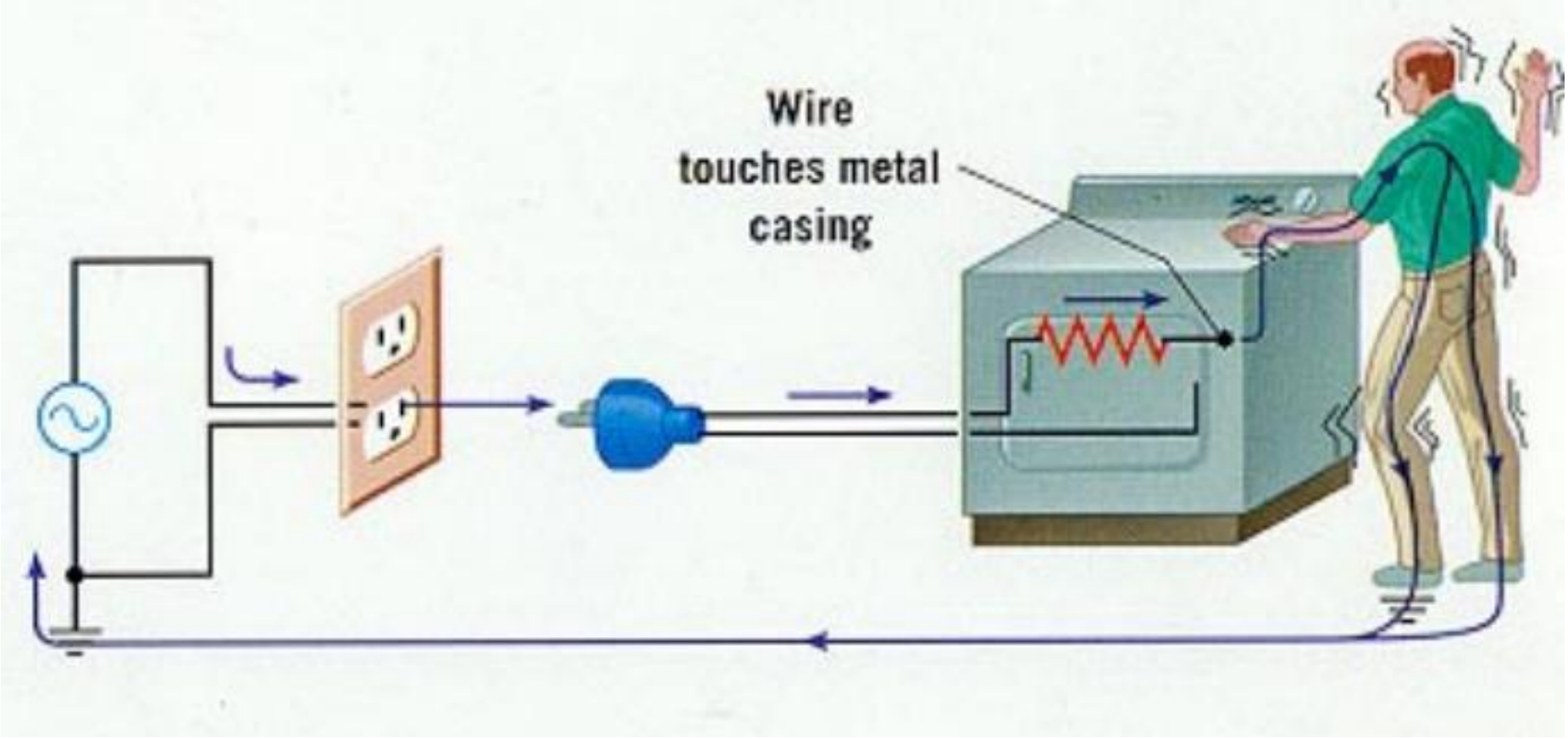


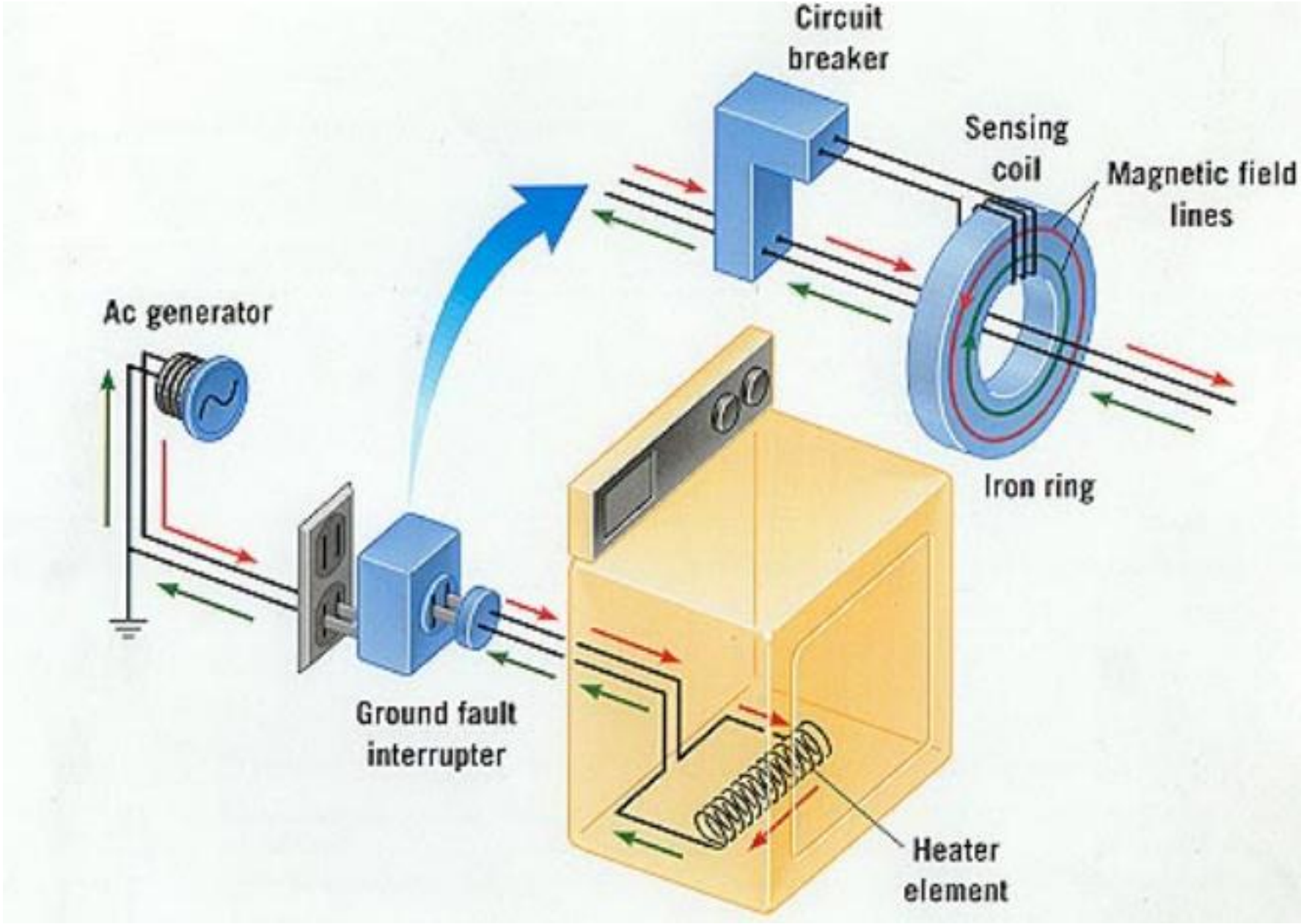

C
S
R
magnetic core
C
S
R
magnetic core

NCSU

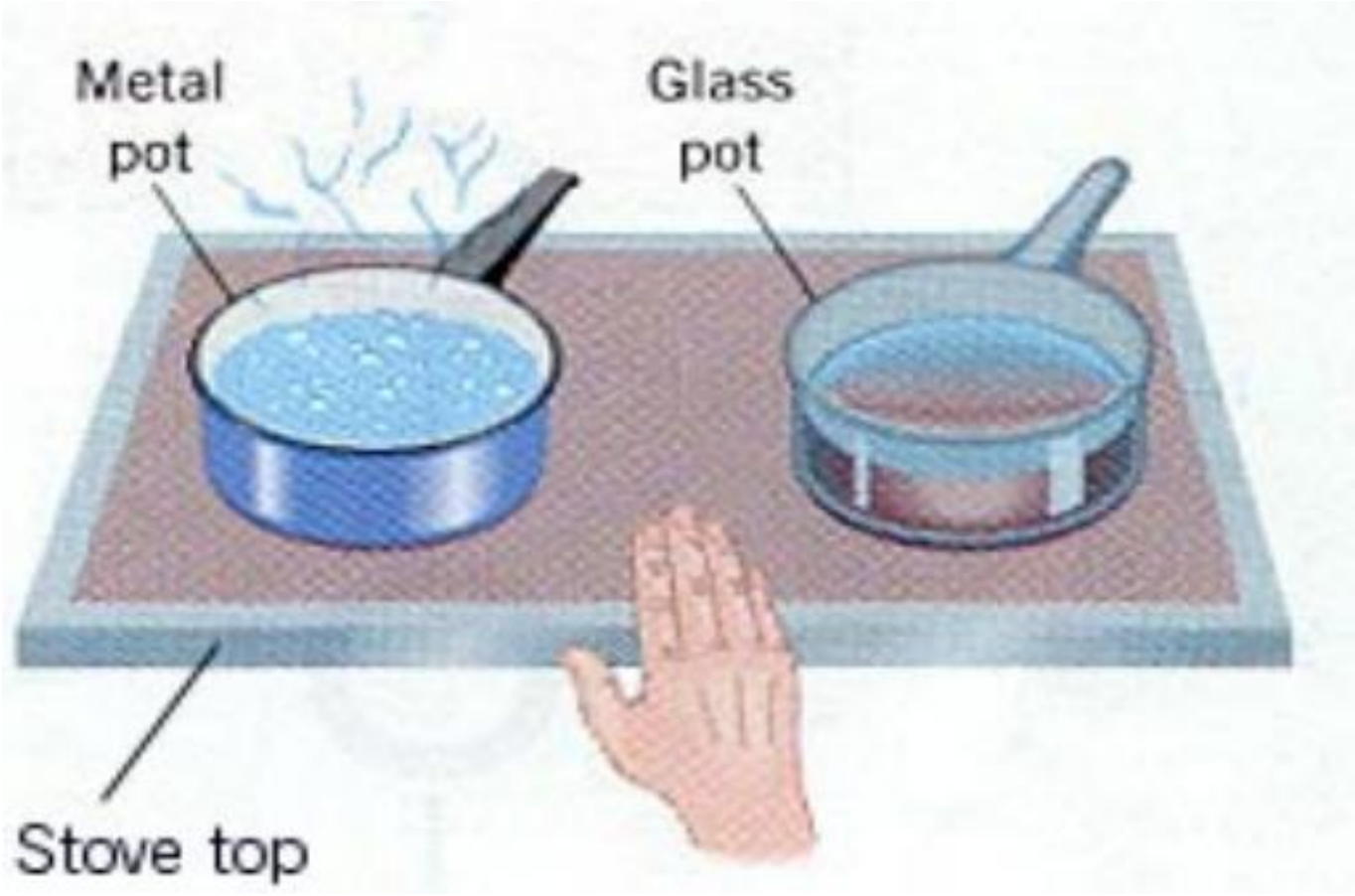
NCSU

2. 漏电保护器



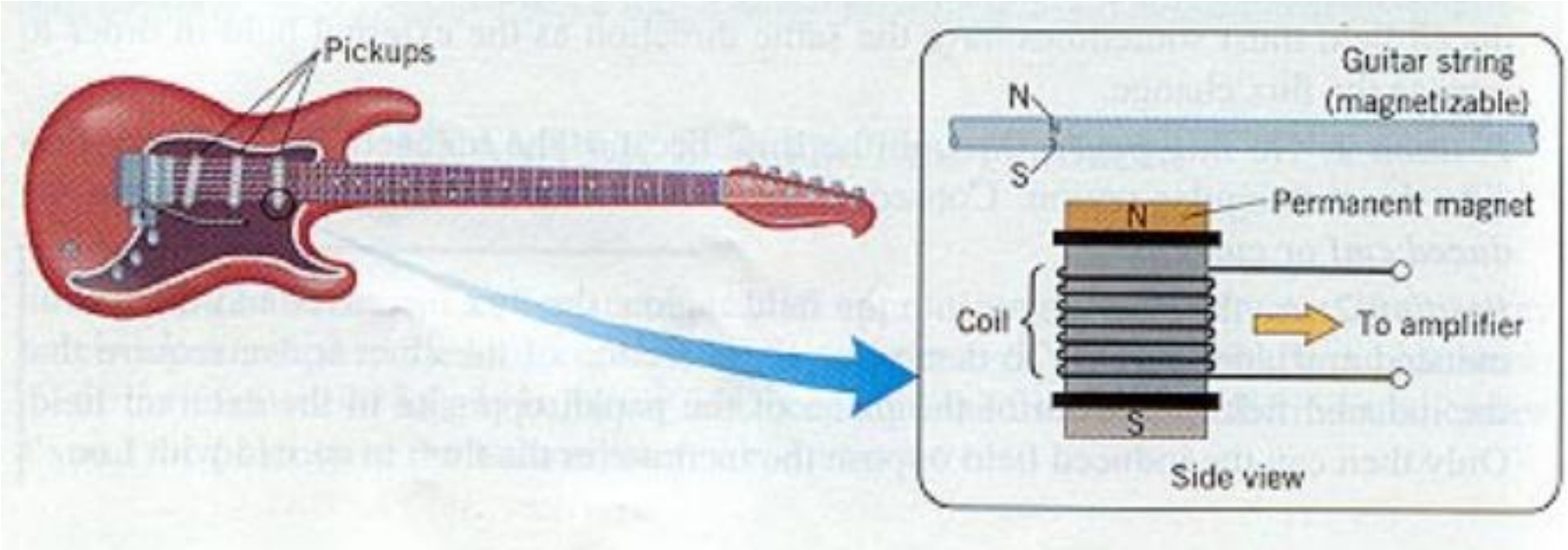

Circuit breaker
Sensing coil
Magnetic field lines
Ac generator
Iron ring
Ground fault interrupter
Heater element

3. 电磁加热炉

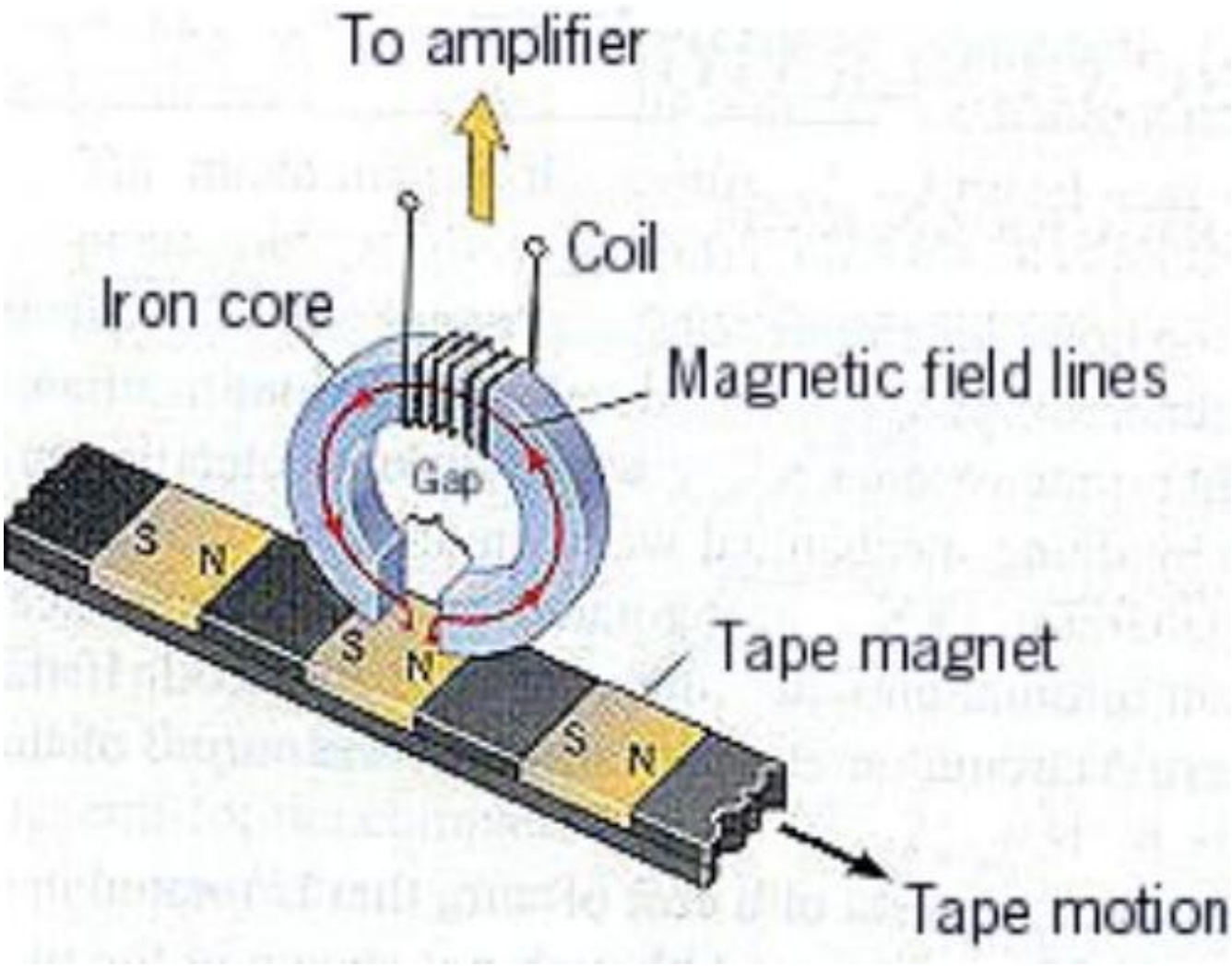


4. 电声设备

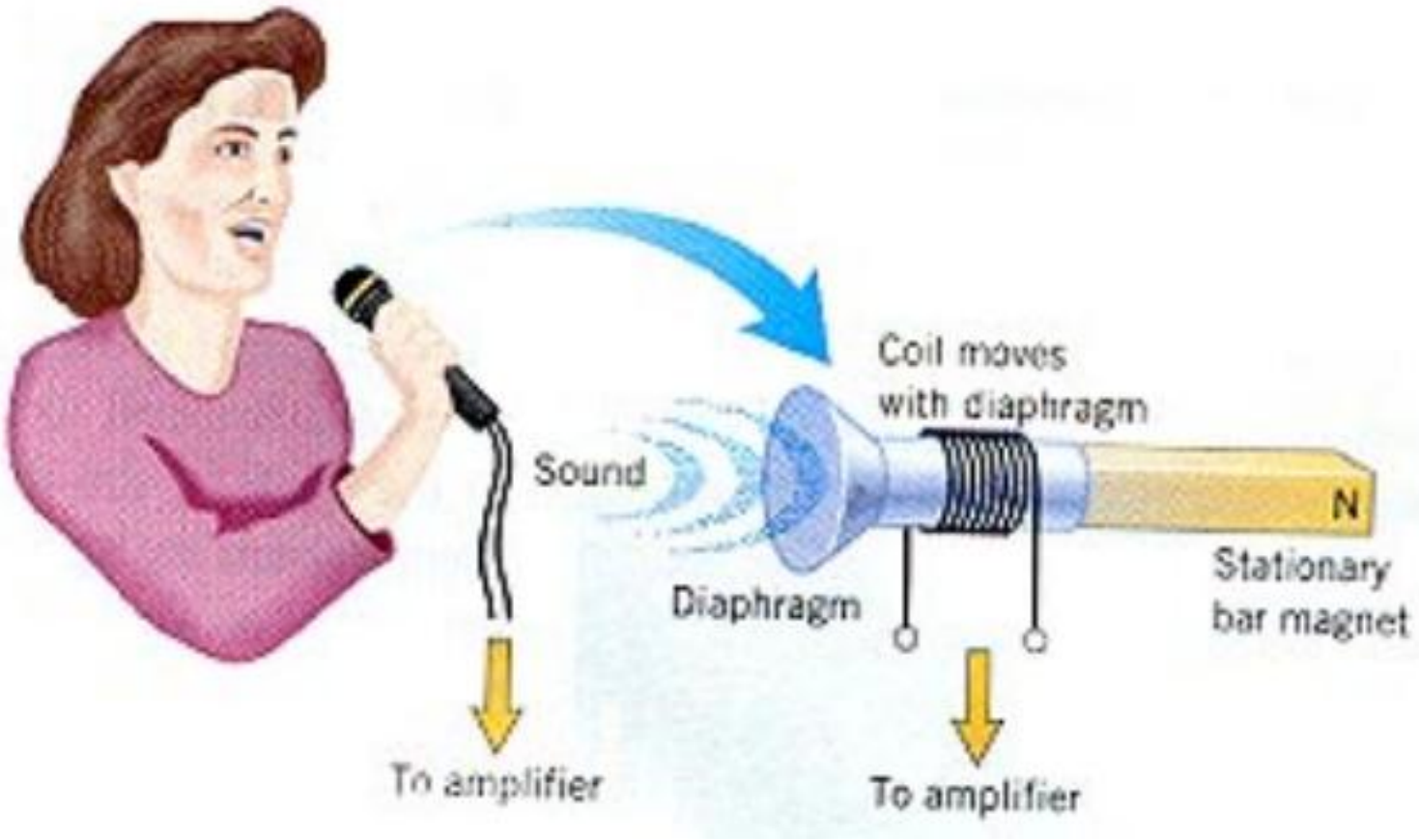
电子吉他


Pickups
Guitar string
(magnetizable)
N
S
Permanent magnet
N
Coil
To amplifier
S
Side view

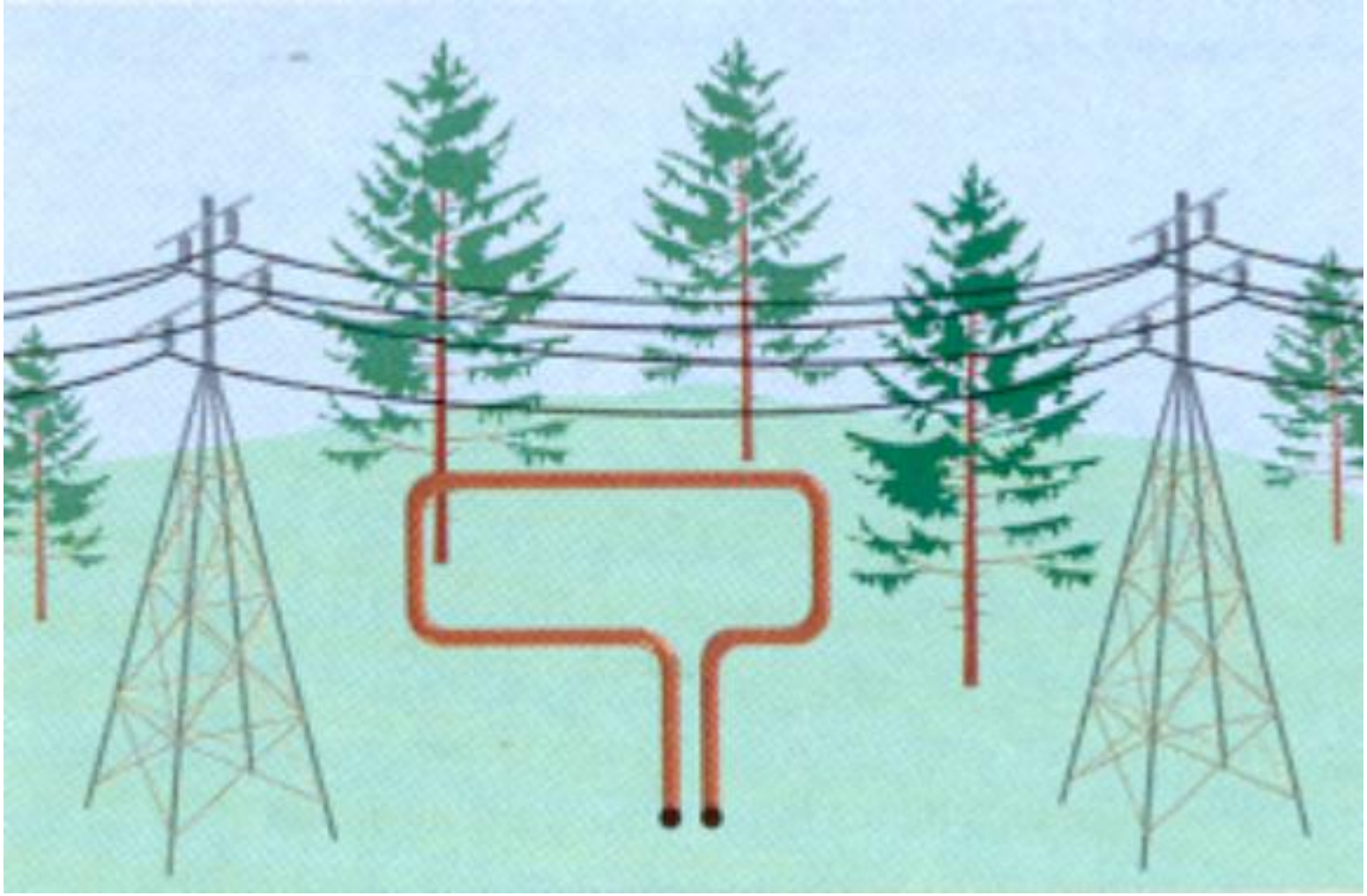
磁头


To amplifier
Coil
Iron core
Magnetic field lines
Gap
S
N
S
N
S
N
Tape magnet
Tape motion

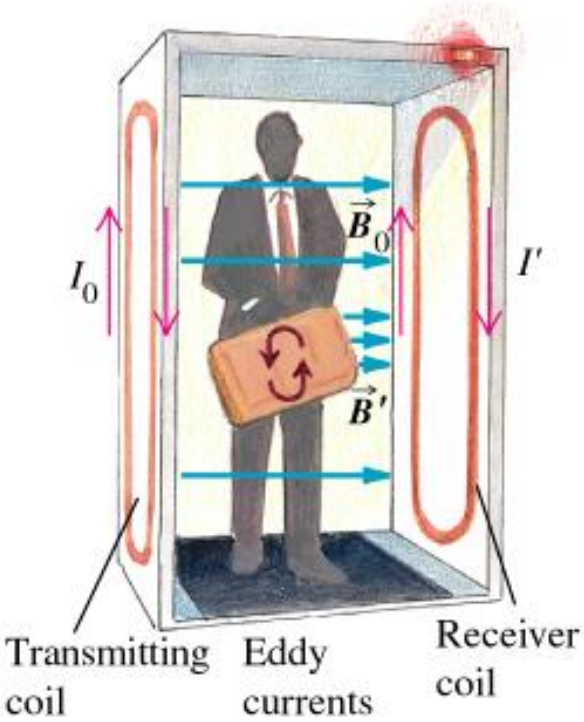
麦克风


Coil moves
with diaphragm
Sound
N
Stationary
bar magnet
Diaphragm
To amplifier
To amplifier

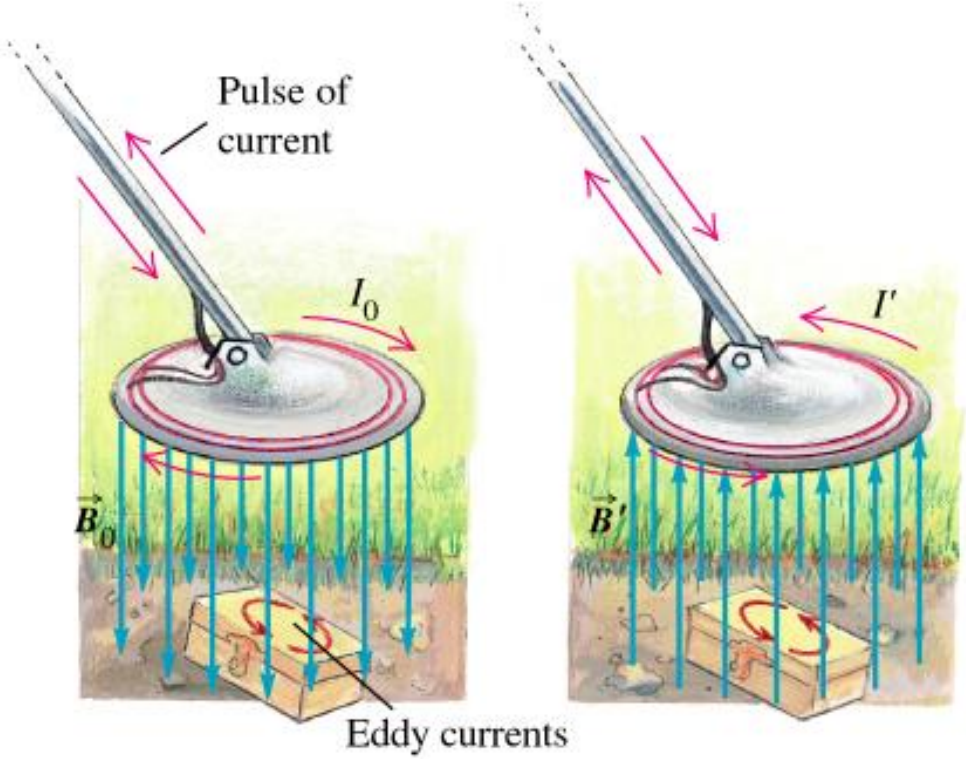
5. Stealing power



6. 其它应用


I_0
$\vec{B}_0$
I'
$\vec{B}'$
Transmitting coil
Eddy currents
Receiver coil

(a)


Pulse of current
I_0
$\vec{B}_0$
I'
$\vec{B}'$
Eddy currents

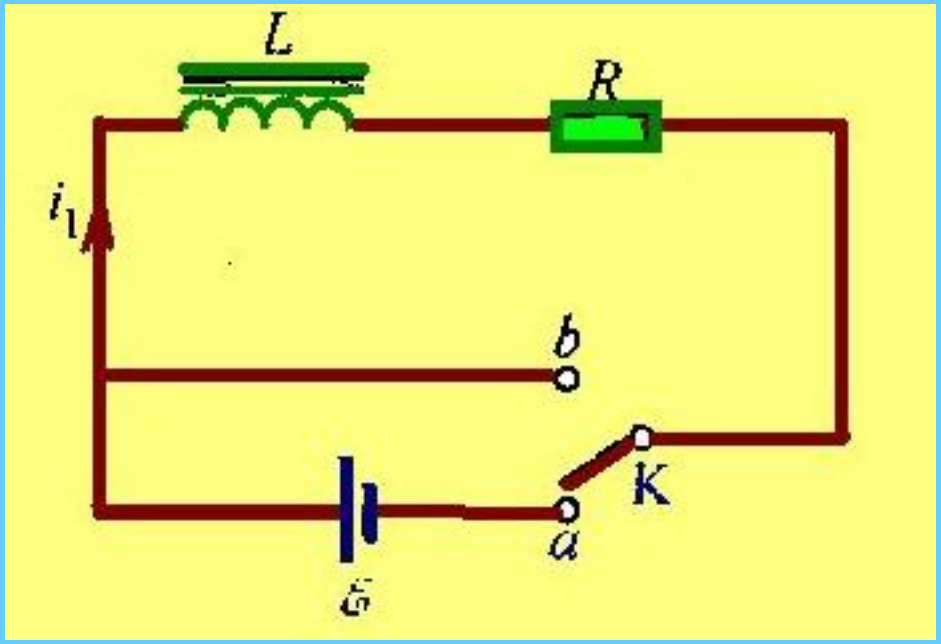
(b)

§6.5 暂态过程

1. RL暂态过程

- 开关在a点时

$$IR = \varepsilon_L + \varepsilon$$



设线圈内阻为零，电感为L

$$IR = -L\frac{dI}{dt} + \varepsilon$$

注意到初始条件，则其解为：

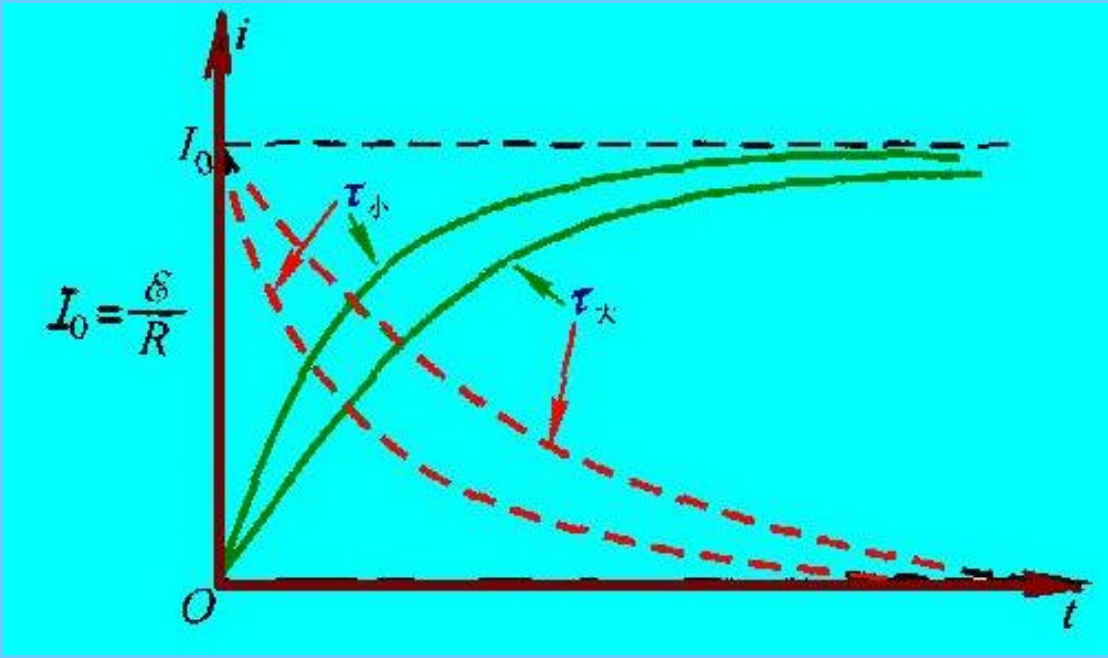
$$I = \frac{\varepsilon}{R}(1 - e^{-\frac{R}{L}t})$$

令 $\frac{\varepsilon}{R}=I_0,\ \frac{L}{R}=\tau$ → $I=I_0(1-e^{-\frac{t}{\tau}})$

当 $t\to\infty, I\to I_0$

当 $t=\tau,\ \mathrm{I}=\mathrm{I}_0(1-e^{-1})=0.63I_0$

τ称**时间常数**，由回路的本身因数确定。一般地，τ只有几个ms或μs。所以称暂态过程。

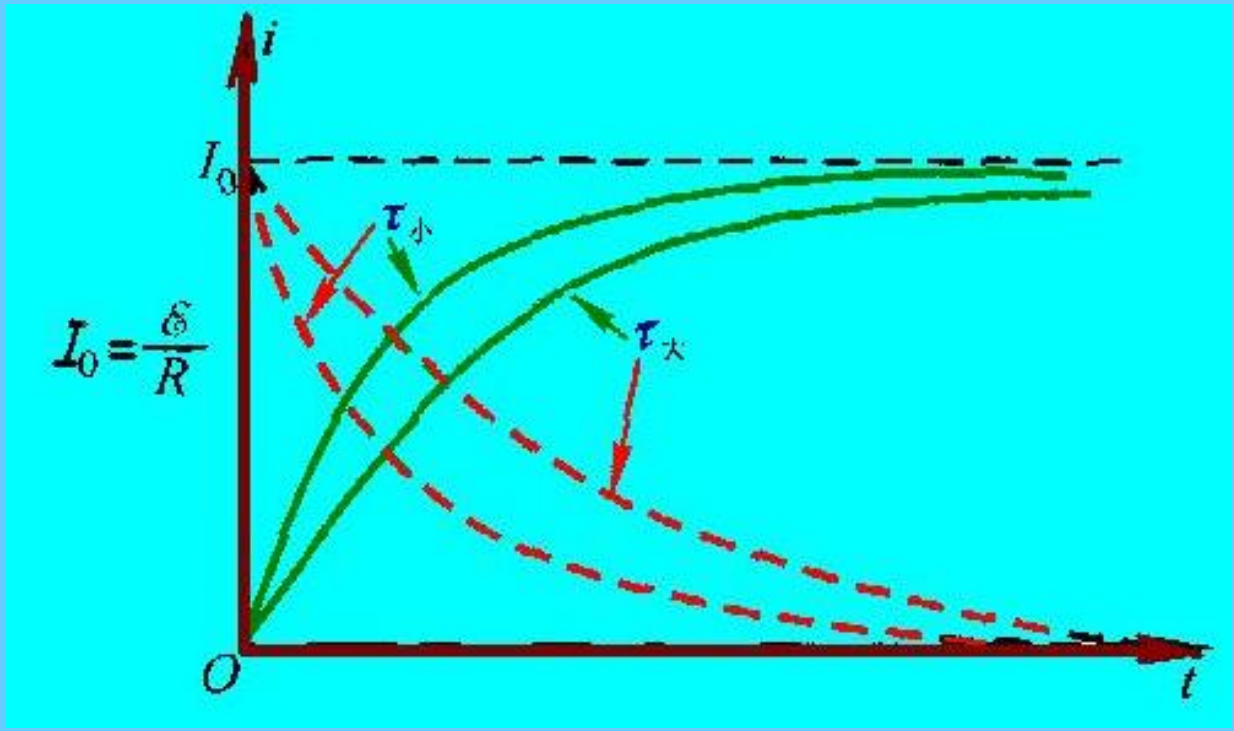


开关在b点时

$$-L\frac{dI}{dt}-IR=0 \quad \Rightarrow \quad I=I_0e^{-\frac{t}{\tau}}$$

当$t\to\infty, I\to 0$

当$t=\tau$, $\mathrm{I}=\mathrm{I}_0e^{-1}=0.37I_0$



2. RC暂态过程

- 开关在a点时

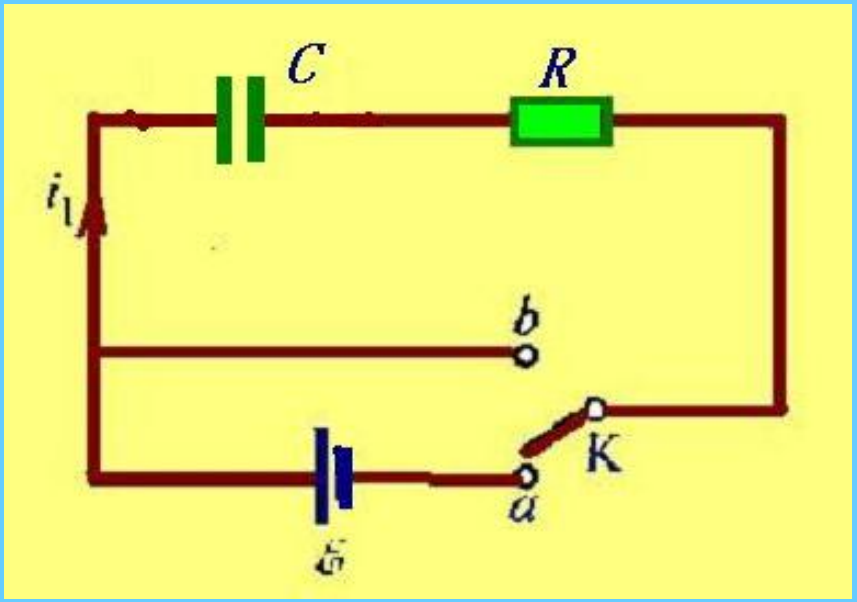
$$\frac{q}{C}+IR=\varepsilon$$

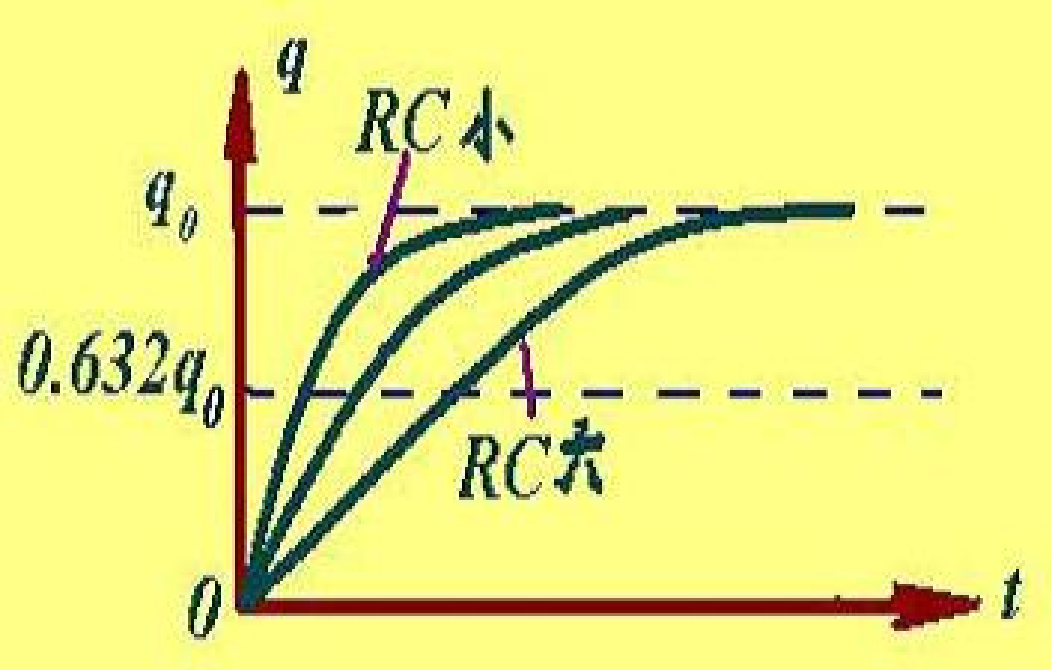
$$R\frac{dq}{dt}+\frac{q}{C}=\varepsilon$$

$$q\big|_{t=0}=0$$

$$q=q_0(1-e^{-\frac{t}{\tau}})$$

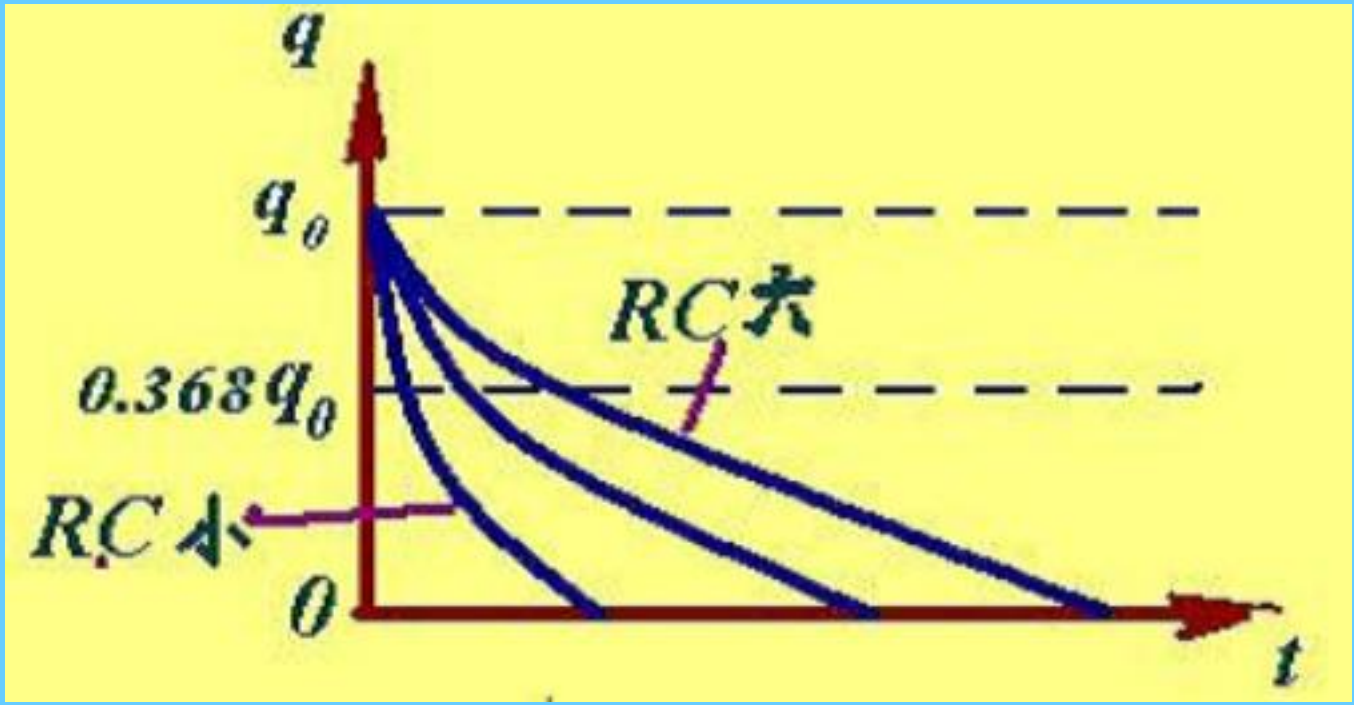
$$q_0=C\varepsilon,\ \tau=RC$$



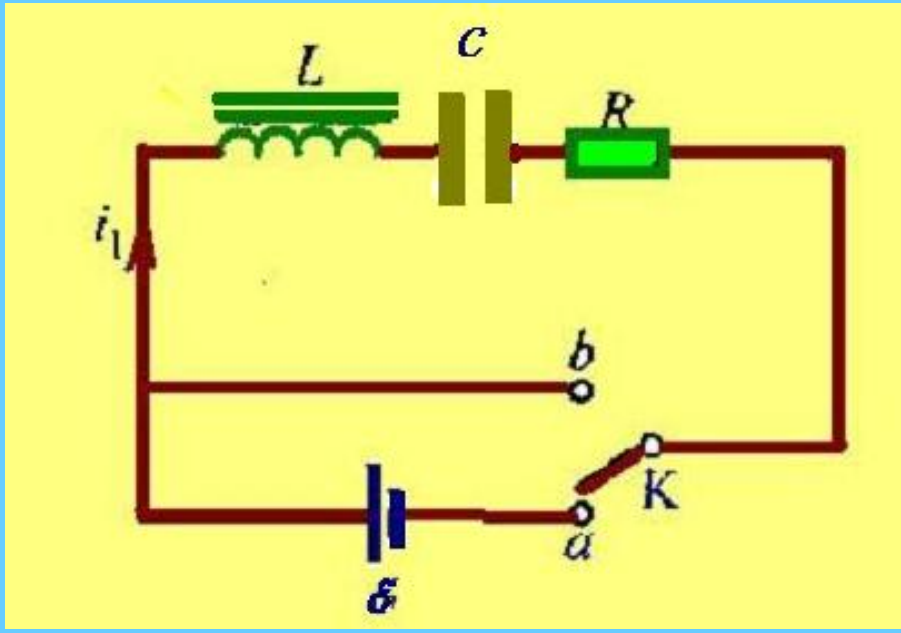


●开关在b点时

$$\begin{cases} R\dfrac{dq}{dt}+\dfrac{1}{C}q=0 \\ q\big|_{t=0}=q_0 \end{cases} \Rightarrow q=q_0e^{-\frac{t}{\tau}}$$



*3. RCL暂态过程



- 开关在a点时

$$L\frac{d^2q}{dt^2}+R\frac{dq}{dt}+\frac{q}{C}=\varepsilon$$

令$\beta=\dfrac{R}{2L},\omega_0=\dfrac{1}{\sqrt{LC}},q_0=C\varepsilon$

$$\frac{d^2q}{dt^2}+2\beta\frac{dq}{dt}+\omega_0^2q=\omega_0^2q_0$$

$$\begin{cases} q\big|_{t=0}=0 \\ \left.\dfrac{dq}{dt}\right|_{t=0}=0 \end{cases}$$

- **欠阻尼** $\beta^2-\omega_0^2<0$

$$q=q_0-q_0e^{-\beta t}(\cos\omega t+\frac{\beta}{\omega}\sin\omega t)$$

$$\omega=\sqrt{\omega_0^2-\beta^2}$$

称阻尼振荡解

- **过阻尼** $\beta^2-\omega_0^2>0$

$$q=q_0-\frac{1}{2\gamma}q_0e^{-\beta t}\left[(\beta+\gamma)e^{\gamma t}-(\beta-\gamma)e^{-\gamma t}\right]$$

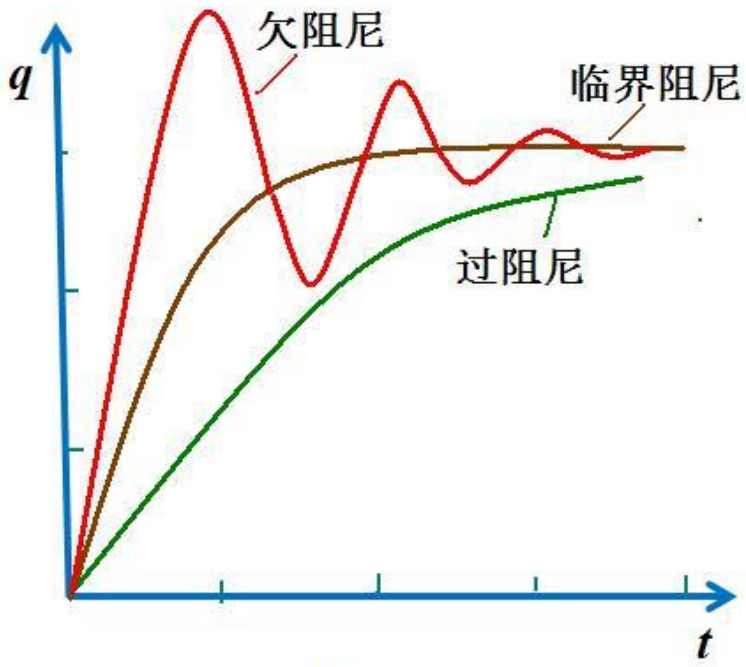
$$\gamma=\sqrt{\beta^2-\omega_0^2}$$

q随时间单调上升，且β越大，上升越慢。

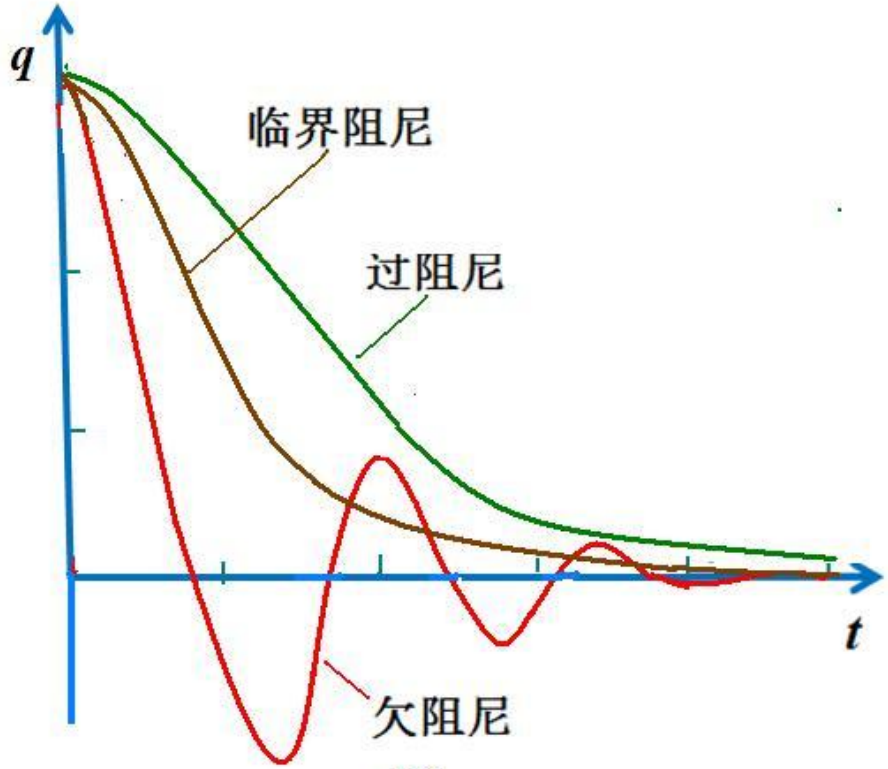
临界阻尼

$$\beta^2 - \omega_0^2 = 0$$

$$q = q_0 - q_0(1+\beta t)e^{-\beta t}$$



(a)



(b)

- 开关在b点时

$$L\frac{d^2q'}{dt^2}+R\frac{dq'}{dt}+\frac{q'}{C}=0$$

$$\begin{cases} q'\big|_{t=0}=q_0 \\ \left.\dfrac{dq'}{dt}\right|_{t=0}=0 \end{cases}$$

请自行讨论其解。

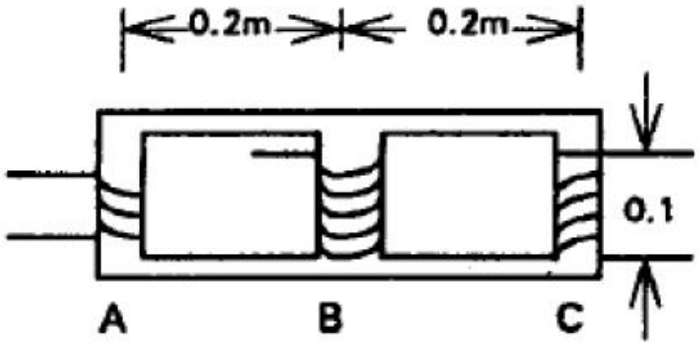
作　业

7.11,　　7.14,　　7.16

7.11 一个变压器如习题 7.11 图所示，线圈 A、B、C 的匝数分别为 500，1000，500，截面积分别是 0.005 m^2，0.001 m^2，0.0005 m^2，铁芯的水平臂截面积是 0.002m^2，如果芯的相对磁导率 $\mu_r = 10000$，求：

(1) 线圈 A 和 C 间的互感；

(2) 线圈 A 和 B 间的互感.



习题 7.11 图

7.14 空心螺线管长 0.5 m，截面 1 cm^2，匝数 1000.忽略边缘效应，它的自感多大？一个 100 匝的副线圈也绕在这个螺线管的中部，互感多大？现有 1A 的稳恒电流流入副线圈，螺线管连接着 10^3 Ω 的负载.如果上述稳恒电流突然停止，将有多少电荷流过电阻？

7.16 由 3×10^6 Ω 的电阻、1 μF 电容和 $\mathcal{E}$ = 4V 的电源连接成简单回路，试求在电路接通后 1s 的时刻，下列各量的变化率：

(1) 电容上电荷增加的速率；

(2) 电容器内储存能量的速率；

(3) 电阻上产生的热功率；

(4) 电源提供的功率.